Endogenous Default Risk and Heterogeneous Credit Transmission*

Pierre Mabille† Olivier Wang‡

December 2025

Why do credit expansions flow to risky borrowers in some markets (e.g., mortgages) but to safe borrowers in others (e.g., credit cards)? We show that two sufficient statistics, the interest-rate elasticity of loan demand and the elasticity of default with respect to loan rates, jointly characterize heterogeneous credit transmission. The feedback between loan contract terms and default risk generates a *credit-risk multiplier* that can amplify or dampen initial shocks depending on demand elasticities. In inelastic markets, credit expansions flow disproportionately to risky borrowers; in elastic markets, the pattern reverses. We use our framework to relate the cross-sectional empirical evidence for the "risk-taking channel" of monetary policy to aggregate bank risk-taking, analyze spillovers and cross-subsidies across borrowers when banks operate on multiple markets. The same statistics determine the impact and persistence of credit crises in a dynamic version of the model.

JEL classification: G21, G28, G33, G51.

Keywords: credit supply shocks, risk-taking channel of monetary policy, subprime borrowers, credit rationing.

---

*We thank Viral Acharya, Arpit Gupta, John Kuong, Alexi Savov, Philipp Schnabl, Alp Simsek, Venky Venkateswaran, and our discussants Laurent Calvet, Sebastian Doerr, William Diamond, Jason Donaldson, Vadim Elenev, Fabrice Tourre, and Quentin Vandeweyer for comments. This paper benefited from participants at the AFA Annual Meeting, the WFA Annual Meeting, the EFA Annual Meeting, the SGF Conference, the CEPR MoFiR Workshop on Banking, the CEBRA Annual Meeting, the CEPR Endless Summer Conference on Financial Intermediation and Corporate Finance, the Banque de France Symposium on Money Banking and Finance, the INSEAD Finance Symposium, the Paris School of Business Workshop on Banks and Financial Markets, and from seminars at NYU Stern, INSEAD, BI Oslo, the Federal Reserve Board, Banque de France, and the Junior European Finance Seminar. We thank the Alfred P. Sloan Foundation and the Macro Financial Modeling (MFM) project for supporting this research.

†INSEAD. Email: pierre.mabille@insead.edu. Website: https://www.pierremabille.com.

‡NYU Stern. Email: olivier.wang@nyu.edu. Website: https://www.olivierwang.com.

1 Introduction

Loan markets are central for the transmission of financial shocks to households and firms. When banks' balance sheets or funding costs move, the terms of credit adjust in ways that are highly heterogeneous across markets and across borrowers with different risk profiles.

Empirically, household credit expansions display very different cross-sectional patterns across loan products. In U.S. mortgage markets, a large literature finds that the pre-crisis boom disproportionately relaxed constraints in riskier segments. Mian and Sufi (2009) show that ZIP codes with initially low credit scores and incomes experienced especially fast mortgage growth. Demyanyk and Van Hemert (2009) and Dell'Ariccia et al. (2012) document a gradual deterioration of loan quality and underwriting standards, and Keys et al. (2010) link securitization to a shift toward riskier mortgages. The common implication is that when funding conditions improve in this segment, the marginal unit of mortgage credit flows toward riskier borrowers or contracts. By contrast, in the credit card market Agarwal et al. (2018) exploit quasi-experimental variation in banks' funding costs and show that most of the additional credit created by a funding shock accrues to high-FICO consumers, with little expansion for low-score cardholders despite their higher marginal propensities to borrow.

Why do credit expansions benefit riskier borrowers in some markets and safer borrowers in others? We take these facts as motivation to build a tractable model of heterogeneous credit transmission across loan markets and borrower types. The central friction is that the probability of default depends on the loan contract: a higher loan rate or loan size make full repayment less likely. This dependence follows the tradition of theoretical credit rationing models since Stiglitz and Weiss (1981) and captures a wide range of microeconomic mechanisms through which higher loan rates or larger loan size make repayment less likely, such as strategic default and adverse selection. Our contribution is to draw general empirical cross-sectional implications from these fundamental frictions.

A key insight of the paper is that the heterogeneous transmission of credit supply shocks is summarized, for each loan market i, by two sufficient statistics: (i) the interest-rate elasticity of loan demand ϵ^i, and (ii) the elasticity of risk with respect to the face value of debt α^i. The demand elasticity ϵ^i measures how strongly quantities respond when the loan rate moves. The default elasticity α^i measures how sensitive credit risk is to the terms of the loan contract. Under weak regularity conditions, and consistent with the empirical literature, borrowers who are riskier in levels (higher default probability) also have higher risk-sensitivity α^i.

We make three main contributions. The first is to characterize analytically how changes in bank funding costs R^f or lending capacity $\bar{L}$ pass through to loan quantities, rates, and default risk in the cross-section. Combining ϵ^i and α^i yields a *credit-risk multiplier* M^i. Intuitively, M^i captures

Inelastic market (e.g., mortgages)	Credit-risk multiplier $M > 1$ $\eta^{\text{Safe}} < \eta^{\text{Risky}} < 1$
Elastic market (e.g., credit cards)	Credit-risk multiplier $M < 1$ $\eta^{\text{Safe}} > \eta^{\text{Risky}} > 1$

Table 1: Pass-through η of credit supply shocks across markets and risk types.

a feedback loop: when funding costs rise, banks raise loan rates, which increases default risk, requiring a further rate increase to compensate lenders. In inelastic markets (i.e., with ϵ^i below some threshold $\bar{\epsilon}$), quantities respond little to rates, so default risk moves in the same direction as rates and the feedback loop amplifies the initial shock, hence the credit-risk multiplier M^i is above 1. In elastic markets (ϵ^i above $\bar{\epsilon}$), the increase in loan rates strongly deters loan demand. The lower loan size helps reduce default risk, hence the feedback loop dampens the shock ($M^i < 1$). In our baseline model and under several standard microfoundations, the threshold that determines the boundary between inelastic and elastic markets is exactly $\bar{\epsilon} = 1$; we also extend the model to allow it to depart from 1.

We then construct pass-through coefficients $\eta^i = M^i \epsilon^i$, which collapse to demand elasticities in the knife-edge case of exogenous default risk ($\alpha^i = 0$). This 2-by-2 mapping between (ϵ^i, α^i) and η^i generates sharp predictions about heterogeneous transmission, summarized in Table 1. In relatively inelastic markets ($\epsilon^i < 1$, empirically consistent with mortgages) pass-through is larger for riskier borrowers (higher α^i), so monetary easing or capital inflows disproportionately expand credit to riskier types. In relatively elastic markets ($\epsilon^i > 1$, empirically consistent with credit cards) the ranking is reversed: safer borrowers are the ones to benefit from credit expansions.

Second, we link these cross-sectional patterns to the total risk in bank portfolios. Any change in loan terms modifies both the composition of bank portfolios and the individual default risk of each borrower. We show that the response of a bank's total portfolio default probability can be decomposed into (i) a within-borrower component, driven by how each μ^i reacts to the induced change in F^i, and (ii) a composition component proportional to $\mathbf{Cov}(\mu^i, \eta^i)$, which captures whether marginal lending flows to riskier or safer borrowers. Our key finding is that these two components have opposite signs. In inelastic markets ($\epsilon^i < 1$), monetary easing reallocates lending toward ex-ante riskier borrowers (which is often interpreted as bank risk-taking), but it simultaneously reduces their individual default probabilities. In elastic markets ($\epsilon^i > 1$), monetary easing increases the share of safer loans but individual default risk rises. Which effect dominates is a quantitative question, but the opposing forces imply that cross-sectional "risk-taking channel" estimates alone may not be sufficient to determine whether aggregate bank risk rises or falls.

Our third contribution is to embed these sufficient statistics into a dynamic model of intermediary net worth following the intermediary asset-pricing literature. We show that the same pass-through coefficient η^i that determines the incidence of shocks in the static model also governs the endogenous impact and persistence of credit crises. For a given shock to bank net worth, higher pass-through makes the lending contraction milder on impact but more persistent, while lower pass-through makes it sharper but shorter-lived. Moreover, when banks are diversified and lend across multiple markets, such as a low-elasticity mortgage market and a high-elasticity credit card market, their common capacity constraint creates cross-market spillovers and dynamic cross-subsidies. For a given shock to total lending capacity, borrowers with a relatively high pass-through experience a starker contraction on impact if their lenders are diversified than if their lenders are specialized, whereas the initial shock is softened for those with a relatively low pass-through. However, over time, credit conditions recover faster for high pass-through borrowers when their bank is diversified, as the presence of the low pass-through borrowers push initial excess spreads upward.

Related literature. Our work contributes to the microeconomic literature on credit markets and to the macro-finance literature on the dynamics of credit crises. On the microeconomic side, we build on the central insight of classic credit-market models that default risk depends on loan terms (Stiglitz and Weiss 1981). Our contribution is to show in a single model that the transnimssion across markets of bank funding and balance-sheet shocks to loan quantities, rates, and default risk can be summarized by two sufficient statistics: the interest-rate elasticity of loan demand ϵ and the elasticity of effective default with respect to the face value of debt α. Their interaction generates a credit-risk multiplier and a market-specific pass-through coefficient.[1]

As in the data, loan terms endogenously depend on both borrower and lender characteristics. Our analysis therefore complements models with heterogeneous households where the interest rate on unsecured consumer loans depends on borrower characteristics (Chatterjee et al. 2007, Livshits et al. 2007), and models of the credit surface where an increase in borrower credit risk leads lenders to tightening collateral requirements (Geanakoplos 2010). We add to these settings by showing that lenders' financial conditions also affect the terms of lending, as in Diamond and Landvoigt (2022) who study the credit surface of borrowers in a rich quantitative model of the mortgage market. In comparison, our contribution is theoretical and analyzes the heterogeneous

[1]In Stiglitz and Weiss (1981), loan size is fixed and lenders adjust only the interest rate, so demand is effectively highly inelastic and the default probability rises steeply with the promised repayment. This environment can be viewed as the extreme case of very low ϵ and high α, which leads to the maximal credit-risk multiplier M.

transmission of bank shocks on *different* loan markets, by analytically explaining how and why loan terms respond given sufficient statistics characterizing the underlying markets. This general formulation delivers a unified analysis of the transmission of bank shocks across different loan markets, and allows us to study the implications for the persistence of credit crises on these markets. Estimates for the elasticities of loan demand and default rates to interest rates are identified in empirical settings which include survey data (Fuster and Zafar 2021), regression discontinuity designs (Fuster and Willen 2017, Best et al. 2019), and structural models (Buchak et al. 2024, Robles-Garcia 2020, Benetton 2021). In our model, we show that they depend on structural parameters which affect borrowers' demand, such as their elasticity of intertemporal substitution, and propensity to default such as moral hazard, adverse selection, and liquidity constraints (Adams, Einav and Levin 2009, Einav, Jenkins and Levin 2012).

A separate strand of work studies how monetary policy and bank funding conditions affect the risk composition of lending. Using large panels of loan-level data, Jimenez et al. (2014), Ioannidou et al. (2014), and Dell'Ariccia et al. (2017) document that lower policy rates or easier funding are associated with a shift in bank portfolios toward ex-ante riskier firms and loans with higher ex-post default rates, a pattern often interpreted as a "risk-taking channel" of monetary policy (related contributions include Borio and Zhu (2012), Bruno and Shin (2015), and Paligorova and Santos (2017), among others). Our characterization shows that such composition effects can be understood through the joint behavior of the demand elasticity ϵ^i and the default elasticity α^i. In relatively inelastic markets, funding shocks naturally tilt lending toward borrowers with higher α^i and higher ex-ante default risk, while in more elastic markets the same shocks can generate the opposite, "flight-to-quality" pattern.

Our dynamic model in Section 5 extends the intermediary asset pricing framework (He and Krishnamurthy 2013, Brunnermeier and Sannikov 2014, Gromb and Vayanos 2018) to credit markets where payoffs are endogenous to loan terms. We show that the same credit-risk multiplier governs the impact and persistence of credit crises: for a given shock to bank capital, higher pass-through makes the crisis milder on impact but more persistent, while lower pass-through makes it sharper but shorter-lived. This mechanism helps understand why credit conditions recover at different speeds across market segments after large disruptions (Justiniano et al. 2019).

Outline. The rest of the paper is organized as follows. Section 2 presents our model of loan contracting with endogenous default risk. Section 3 derives our main results on heterogeneous credit transmission, and Section 4 draws implications for bank risk-taking, decomposing aggregate risk into within-borrower and composition effects. Section 5 embeds our framework into a dynamic model and shows how pass-through also characterizes the impact and persistence of credit crises.

Section 6 extends the baseline model to more general default functions and intensive-margin rationing.

2 A model of competition in loan contracts

We develop a simple model of equilibrium loan contracting subject to financial frictions that is designed to nest several standard environments.

2.1 Environment

We consider a set of $B \geq 2$ identical lenders or "banks" indexed by $b = 1, \ldots, B$, and heterogeneous borrowers indexed by their observable type $i = 1, \ldots, I$, with $I \geq 1$. There is a mass $N^i > 0$ of type-i borrowers.

Throughout, we interpret each borrower type i as a "market" or segment of the loan market defined by observable characteristics that are common within the segment, such as product type (e.g. mortgages vs. credit cards), collateral category, geography, or a range of credit scores. This parsimonious formulation allows both the demand curve and the risk to differ across markets, so that, for instance, prime mortgages, subprime mortgages, and small-business loans can all be represented by different indices i.

In the baseline we assume that every bank can lend to all markets with the same technology and funding cost. Section 3.2 then contrasts specialization and diversification, by letting banks be active on different subsets of markets. When diversified banks lend across multiple markets, their common lending constraint creates cross-market spillovers and cross-subsidies in prices and quantities relative to the case of bank specialization. This maps to recent evidence on segmentation across products and geographies (e.g., Paravisini et al. 2023, Blickle et al. 2023).

Loan contracts and default risk. We denote loan contracts $C^i = \left(R^i, l^i\right)$. Contract terms depend on the observable borrower type i, and include two dimensions: an interest rate R^i and a loan amount l^i that the borrower receives at origination. The face value $F^i = R^i l^i$ corresponds to the repayment absent default. Accounting for credit risk, the expected payoff from a loan C^i is

$$\underbrace{R^i l^i}_{\text{without default}} - \underbrace{\mu^i(C^i) R^i l^i}_{\text{expected loss}} \tag{1}$$

The term $\mu^i(C^i) \in [0, 1]$ is the *effective default probability*, defined as the bank's expected loss per unit of face value $F^i = R^i l^i$. If the lender recovers nothing upon default, then μ^i is exactly the default probability, whereas μ^i is below the actual default probability when there is a positive recovery rate upon default (equivalently, the loss given default lower than the full face value F^i).

Crucially, the effective default probability μ^i may depend on the loan contract terms C^i. As in the credit rationing literature, we seek to capture how a higher rate R^i or a larger loan size l^i make it harder for the borrower to repay the loan in full. Using the effective default probability μ^i as a primitive object allows us to isolate the relevant properties of μ^i that do not depend on the specific underlying foundations while nesting several environments in which ex-ante or ex-post information frictions make the expected loss depend on loan terms. In particular, it allows us to treat as equivalent a wide variety of environments in which higher promised payments make full repayment less likely: liquidity default driven by low realized income, strategic default driven by low collateral values, and adverse selection driven by a worse composition of borrowers selecting into a given contract. All that matters for our analysis is how the expected loss per unit of face value reacts to F^i. In Appendix A, we show how different microeconomic frictions map into μ^i in these three leading examples of endogenous default risk.

Since any payoff can be decomposed as (1), non-trivial implications can only stem from additional restrictions on the function $\mu^i(C^i)$. We make the following assumption throughout the paper to discipline how μ^i depends on contract terms C^i:

Assumption 1 (Endogenous default risk)**.** *The effective default probability of a type-i borrower under a loan contract $C^i = (R^i, l^i)$ is given by a function $\mu^i(F^i)$ that only depends on the face value $F^i = R^i l^i$.*

The interpretation of Assumption 1 is that the relevant margin for credit risk is the total repayment burden F_i, not its decomposition into principal and interest. This is natural when the borrower's default status is determined ex post, after receiving the loan proceeds. It is implied by many static settings in which the distribution of borrower income is given, including the ones we detail in Appendix A.

Remark 1 (Relaxing Assumption 1)**.** *Our results extend qualitatively to richer models in which μ^i is allowed to depend separately on R^i and l^i, for instance because the initial loan size l^i shifts the distribution of future income earned by the borrower, which can then affect credit risk even holding F^i fixed. The general point is that the incidence of credit supply shocks is qualitatively different in markets with low vs. high demand elasticity. While Assumption 1 allows us to obtain simple characterizations in terms of an elasticity threshold exactly equal to 1, we show in Section 6.1 how the same forces would still lead to differences in transmission between low- and high-elasticity markets in more general formulations of μ^i. The main difference is that the elasticity threshold $\bar{\epsilon}$ that defines the boundary between "inelastic" ($\epsilon < \bar{\epsilon}$) and "elastic" ($\epsilon > \bar{\epsilon}$) markets can depart from 1. The general expression $\bar{\epsilon} = \frac{R\mu_R}{l\mu_l}$ simplifies to 1 under Assumption 1.*

For each borrower type i, we define the *default elasticity*

$$\alpha^i = -\frac{\partial \log(1-\mu^i)}{\partial \log F^i}, \tag{2}$$

which captures how credit risk μ^i varies with the face value of the debt F^i. When $\alpha^i = 0$, credit risk is exogenous. Default risk can be high, as captured by a high μ^i, but it does not vary with the terms of the loan. When $\alpha^i > 0$, however, credit risk increases with the interest rate and the quantity borrowed.

We restrict attention to default elasticities satisfying $0 \leq \alpha^i < 1$. Equivalently, this means the loan's payoff (1) increases with the interest rate. Seminal papers on credit rationing (e.g., Stiglitz and Weiss 1981, Williamson 1987) show that a sufficiently strong dependence of credit risk in the interest rate (i.e., α^i being higher than 1) can cause profits to be backward-bending in the loan rate R^i. This can in turn induce lenders to ration credit at the extensive margin, by denying credit altogether to some borrowers even though they would be willing to pay a rate above the equilibrium rate.[2] We will show that many empirically relevant implications of endogenous default risk arise even when abstracting from such extreme perverse "Laffer curve" effects, or alternatively even when restricting attention to the left-side of the Laffer curve over which payoffs are *not* backward-bending.

Borrowers' credit demand. Borrowers are characterized by their indirect utility over loan contracts $V^i(R, l)$. This formulation can nest both households' and firms' optimization problems. For households, V^i is the outcome of utility maximization driven by consumption smoothing and insurance motives. For firms, V^i captures incentives to invest and hedge under financial constraints.

We denote $\ell^i(R^i)$ borrowers' unconstrained loan demand given a rate R^i, defined as solving the standard Euler equation

$$V^i_l(R^i, l^i) = 0. \tag{3}$$

For each borrower type i, we define the interest-rate elasticity, or simply demand elasticity, as

$$\epsilon^i = -\frac{R^i}{\ell^i}\frac{d\ell^i}{dR^i}. \tag{4}$$

The pairs (α^i, ϵ^i) will be the central statistics summarizing the heterogeneous transmission of credit shocks.

We impose standard regularity conditions on borrowers' indirect utility:

[2]We return to the connection to these models when introducing the credit-risk multiplier in Section 3.1.

Assumption 2. *For each borrower type i, the indirect utility V^i is differentiable, decreasing in the loan rate, $\frac{\partial V^i}{\partial R} < 0$, concave in the loan size, $\frac{\partial^2 V^i}{\partial l^2} \leq 0$, and the marginal cost of a higher rate is higher at a larger debt burden: $\frac{\partial^2 V^i}{\partial R \partial l} \leq 0$.*

Assumption 2 is natural in a wide range of settings, including the examples in Appendix A. The last part states that the loan demand curve $\ell^i(R^i)$ is downward-sloping.

The *credit wedge* associated with a loan contract C^i, is defined as $\tau^i(C^i) = -\frac{l^i V_l^i}{R^i V_R^i}$ and measures how constrained type-i borrowers are if they accept the loan contract C^i. In the baseline, we impose $\tau^i = 0$, so loan size is pinned down by the borrower's demand $l^i = \ell^i(R^i)$ at the posted rate. When $\tau^i > 0$ is allowed (Section 6.2), a positive wedge means that the contract features credit rationing at the intensive margin: borrowers would prefer to borrow more at the prevailing rate, but are prevented from doing so by a binding quantity restriction embedded in the contract.

Contracts are exclusive, in the sense that borrowers can only accept one contract. We focus on settings in which borrowers indeed get positive utility from borrowing.[3]

Borrowers optimally choose which set of contracts is acceptable: given the set of all available contracts offered by banks $\{C_b^i\}_{b=1,\ldots,B}$, they set acceptance probabilities $\{q_b^i\}_{b=1,\ldots,B}$ for each bank such that $\sum_{b=1}^B q_b^i = 1$.

Let $V_{\min}^i$ denote borrower i's outside option value (e.g., not borrowing). We focus on equilibria with participation, so offered contracts satisfy $V^i(C) \geq V_{\min}^i$. A bank b only enters borrower i's best response set, BR^i, if it offers a contract that is at least as attractive as the best outside option available from other lenders $b' \neq b$. If the borrower is indifferent between several banks, then she randomly accepts one of them with uniform probability. If bank b offers a strictly more attractive contract than all other lenders, then borrower i only accepts bank b's contract. Formally, denote for each borrower type i and bank b

$$\text{BR}_{-b}^i = \arg\max_{b' \neq b} V^i(C_{b'}^i), \quad \overline{V}_{-b}^i = \max_{b' \neq b} V^i(C_{b'}^i), \tag{5}$$

where BR_{-b}^i is the subset of banks other than b posting the contracts that borrower i finds most attractive, and $\overline{V}_{-b}^i$ is the highest expected value that borrower i can obtain outside bank b. The loan application strategy of borrower i is described by the probability that she accepts bank b's contract, which is given by

$$q_b^i(C_b^i, C_{-b}^i) = \begin{cases} 1 & \text{if } V^i(C_b^i) > \overline{V}_{-b}^i \\ \frac{1}{|\text{BR}_{-b}^i|+1} & \text{if } V^i(C_b^i) = \overline{V}_{-b}^i \\ 0 & \text{if } V^i(C_b^i) < \overline{V}_{-b}^i \end{cases} \tag{6}$$

[3]Alternatively, we could label no-borrowing as $b = 0$ at the cost of additional notation.

where we denote $C^i_{-b} = \left\{C^i_{b'}\right\}_{b' \neq b}$ the set of contracts offered by banks other than b, and $\left|\mathrm{BR}^i_{-b}\right| \in \{0, 1, \ldots, B-1\}$ is the number of banks in the set BR^i_{-b}, with $\left|\mathrm{BR}^i_{-b}\right| = 0$ if $\overline{V}^i_{-b} < V^i_{\min}$.

Banks' credit supply. In our baseline model, banks b are risk-neutral, perfectly competitive, and subject to capacity constraints on lending. We first assume that all banks have "access" to all types of borrowers, that is, there is no segmentation; we later discuss implications of more subtle patterns in bank specialization (Paravisini et al., 2023; Blickle et al., 2023). For each borrower type i, bank b posts a contract C^i_b. The expected profit from making a loan C^i_b is

$$\Pi^i(C^i_b) = \underbrace{R^i l^i}_{\text{without default}} - \underbrace{\mu^i(C^i) R^i l^i}_{\text{expected loss}} - \underbrace{R^f l^i}_{\text{funding cost}} \tag{7}$$

where R^f is the bank's funding cost, assumed identical across banks.

The bank also chooses a number $n^i_b \geq 0$ of loans to make to borrowers of type i. Banks are subject to a capacity constraint on their lending, which we write in general form as:

$$\sum_{i=1}^{I} \rho^i n^i_b q^i_b l^i_b \leq \overline{L}. \tag{8}$$

For each borrower type i, the bank lends out a total amount $n^i_b q^i_b l^i$, hence the left-hand side of (8) represents the bank's weighted total lending, which must be lower than the lending capacity $\overline{L}$.

The parameters $\rho^i \geq 0$ are risk-weights that can be interpreted as regulatory risk-weights, for instance if lending to some (presumably riskier) types of borrowers requires more regulatory capital. They can also capture differences in banks' ability to securitize different types of loans and take it off their balance sheets (e.g., conforming vs. non-conforming mortgages). Unless noted otherwise, we abstract from differences in risk-weights and assume $\rho^i = 1$ for all borrower types i; we only discuss the implications of differences in ρ^i in Section 3.2.

The bank lending capacity $\overline{L}$ can arise from regulatory constraints such as capital requirements and from market-based constraints imposed by bank creditors due to informational issues similar to those affecting the bank-borrower relationship. A large literature (e.g., Holmstrom and Tirole 1997, Gertler and Kiyotaki 2010) provides microfoundations for the moral hazard or limited commitment problems that lead banks themselves to face credit constraints. We start by taking shocks to $\overline{L}$ as given, to focus on how credit supply shocks are transmitted to different borrowers through the multiple terms of loans. In Section 5, we embed our loan contracting model into a dynamic framework that relates the path of lending capacity over time $\left\{\overline{L}_t\right\}_t$ to the dynamics of banks' net worth, and thus profits from past loans.

Bank optimization problem. Each bank b takes other banks' contracts $C^i_{-b} = \left\{C^i_{b'}\right\}_{b' \neq b}$ and borrowers' acceptance strategies $q_b^i(\cdot, C_{-b}^i)$ as given, and chooses contracts $\{C_b^i\}^i$ and quantities $\{n_b^i\}^i$ to maximize total expected profits across borrowers

$$\sum_{i=1}^{I} n_b^i q_b^i \Pi^i \left(C_b^i\right) \tag{9}$$

subject to the capacity constraint (8). In the baseline model, we assume that banks can only offer contract without rationing, that is satisfying the no-rationing constraint:

$$l^i = \ell^i(R^i). \tag{10}$$

Section 6.2 shows that allowing intensive-margin rationing $l^i < \ell^i(R^i)$, i.e., contracts with loan size strictly below borrowers' desired loan size at the prevailing rate, does not change the main results.

In the total expected profits function, the probability q_b^i to attract a borrower depends on her contract C_b^i and on her other options $\{C_{b'}^i\}_{b' \neq b}$ according to the borrower's optimal strategy (6). This implies that banks take into account borrowers' optimization across contracts offered by different banks, in the same way as firms take consumers' product choice into account in Bertrand-Nash competition.

The quantity of loans n_b^i captures potential credit rationing at the *extensive* margin. Importantly, even though loan volume enters the bank capacity constraint (8) only through the product of the number of loans with the individual loan size ($n_b^i l_b^i$), in general the two dimensions n_b^i and l_b^i are not equivalent from the bank's viewpoint: making $n_b^i = 10$ loans with individual loan size $l_b^i = \$10$ is not the same as making a single ($n_b^i = 1$) loan of size $l_b^i = \$100$. Consequently, credit rationing at the extensive and intensive margins have different implications. Formally, the total expected profits from lending to a borrower, $n_b^i \Pi^i \left(C_b^i\right) = n_b^i l_b^i \left[R_b^i(1 - \mu^i(C_b^i)) - R^f\right]$, cannot be written as a function of the product $n_b^i l_b^i$ when individual loan size l_b^i affects the effective default probability μ^i.

Solving the optimization problem of the bank may appear as challenging because small deviations in the contract offered by the bank can cause jumps in borrower's acceptance rate q_b^i according to (6). However, it turns out that the extensive margin variable n_b^i can circumvent this difficulty. The reason is that from the bank's perspective, any response of borrowers' demand to contract terms can be "undone" through the proper choice of n_b^i. Intuitively, given individual contract terms C_b^i, the same expected number of loans can be attained by reaching out to more borrowers (higher n_b^i) or converting more offers (higher q_b^i). More precisely, as long as the bank offers a contract C_b^i that satisfies type i borrowers' participation constraint (11)

$$V^i \left(C_b^i\right) \geq \overline{V}_{-b}^i, \tag{11}$$

then q_b^i is strictly positive (equal to either $1/B$ or 1). Therefore, the bank's problem can be rewritten as maximizing the following objective:

$$\sum_{i=1}^{I} X_b^i l_b^i \left\{ R_b^i \left[1 - \mu^i(R_b^i l_b^i)\right] - R^f \right\} \tag{12}$$

subject to the lending capacity constraint $\sum_{i=1}^{I} X_b^i l_b^i \leq \overline{L}$ where $X_b^i = n_b^i q_b^i$ denotes the expected number of loans to borrower type i. Moreover, it is without loss of generality to restrict attention to contracts that satisfy (11), because the bank could always choose to deny credit to borrowers of type i by setting $n_b^i = 0$ instead of offering a contract that does not satisfy (11) and is therefore never accepted ($q_b^i = 0$).

2.2 Examples

Minimal example: Strategic default with Pareto distribution. In what follows, we sometimes refer to the following concrete example to explain key features of the equilibrium. A borrower (a firm or a household) faces no income risk, but needs to pledge collateral with a stochastic realized private value v. "Strategic default" happens if the realization of v is too low and the borrower prefers defaulting on the debt and losing the collateral. A higher repayment value $F = Rl$ makes repayment less likely, because there are more states of the world in which the benefit from defaulting exceeds the cost in terms of foregone collateral.

Borrowers face a deterministic income path y_0, y_1 to abstract from precautionary saving effects, but each borrower pledges collateral whose private value v is a random variable v realized at $t = 1$, with cumulative distribution function G and density $g = G'$. A borrower owing debt F repays if and only if

$$u(y_1 - F) + v \geq u(y_1) \Leftrightarrow v \geq \hat{v}^*(F) \equiv u(y_1) - u(y_1 - F), \tag{13}$$

where the optimal default threshold $\hat{v}^*$ is an increasing function of the face value F. Lenders' recovery rate is zero in case of default, hence the effective default probability is just the actual default probability $\mu^i(F) = \mathbf{P}^i\left[v \leq \hat{v}^*(F)\right]$.

While in general the default elasticity (2) depends on the face value F^i where it is evaluated, a tractable special case obtains when the variable $(\hat{v}^*)^{-1}(v)$ is distributed according to a Pareto distribution with shape parameter α:

$$\mu^i(F) = \mathbf{1}[F \geq F_{\min}] \times \left[1 - (F_{\min}/F)^{\alpha}\right]. \tag{14}$$

where $\alpha, F_{\min} > 0$. This functional form is particularly convenient because it leads to a default elasticity equal to the Pareto parameter α, independently of the loan contract.

Other examples. Besides the strategic-default example, Appendix A presents two additional microfoundations that give rise to a default function μ^i satisfying our assumptions.

In a "liquidity default" environment, borrowers face stochastic income at $t = 1$ and must repay their debt out of this risky cash flow. Default occurs whenever realized income falls below a threshold that depends on the face value F (for example, when income net of a subsistence level is insufficient to cover the promised payment). The default probability is then the lower tail of the income distribution evaluated at a cutoff that is increasing in F so that both the level and the elasticity of default are determined by the shape of that tail. For instance, for log-concave distribution H over date-1 income and default, $\alpha(F)$ increases with the baseline risk $\mu(F) = H(F)$.

In an "adverse selection" environment, borrower types differ in their default probabilities (which can each be exogenous), but the bank observes only the contract terms. For any given interest rate and loan size, only borrowers whose expected surplus is positive choose to borrow, so a larger face value F both raises the default probability of a given type and worsens the composition of types that select into the contract. Aggregating across types yields an effective default probability μ that is again increasing and typically convex in F.

While the baseline interpretation of the model is static, Appendix C.1 shows how the framework can be reinterpreted to allow for a dynamic repayment schedule.

These examples illustrate that Assumption 1 is satisfied in environments with very different underlying frictions (risky collateral values, risky income, and asymmetric information) so that the reduced-form statistics we use below capture a wide class of settings with endogenous default risk.

2.3 Equilibrium

Our concept of equilibrium corresponds to Bertrand-Nash competition in loan contracts:

Definition 1. *An equilibrium is a borrower strategy $\{q_b^i(\cdot)\}_b$ for each borrower type i and a bank strategy $\{n_b^i, C_b^i\}^i$ for each bank b such that:*

(i) given all the available contracts $\{C_b^i\}_b$, borrowers optimize according to (6);

(ii) given borrower strategies (6), banks maximize (9) subject to (8), (10), and (11);

(iii) for all borrower types i, markets clear: $\sum_b n_b^i q_b^i = N^i$.

We focus on symmetric equilibria, henceforth referred to as equilibria, such that all banks b offer the same contract $C_b^i = C^i$ and choose the same quantity $n_b^i = n^i$ and we can thus omit b subscripts. Therefore, in equilibrium, all borrowers accept any of the contracts offered with probability $q_b^i = 1/B$ for all i, b, hence the market clearing condition for each borrower type i simplifies to $n^i = N^i$.

Our first main result characterizes the equilibrium allocation when banks face a lending capacity constraint, and shows how constrained banks allocate scarce balance sheet capacity across borrower types.

Proposition 1. *In any equilibrium,*

i. ***Optimal allocation across borrower types:*** *Banks equalize the profit per dollar across borrower types, setting*

$$\frac{\Pi^i}{l^i} = R^i \left[1 - \mu^i(R^i l^i)\right] - R^f = \nu \tag{15}$$

for all borrower types i, where $\nu \geq 0$ is the multiplier on the bank's capacity constraint.

ii. ***Equilibrium characterization:*** *For each borrower type i, denote the unconstrained contract $C_*^i = \left(R_*^i, l_*^i\right)$ the solution to*

$$\Pi^i\left(\ell^i(R^i), R^i\right) = 0. \tag{16}$$

If the unconstrained loan quantities l_*^i satisfy the capacity constraint (8), that is:

$$\sum_{i=1}^{I} \frac{N^i}{B} l_*^i \leq \overline{L}, \tag{17}$$

then banks are unconstrained in the sense that (8) is slack in equilibrium. In that case, the equilibrium contracts are given by C_*^i and banks earn zero profit ($\Pi^i = 0$) on all borrowers.

Conversely, if (17) does not hold, then banks are constrained in the sense that (8) binds in equilibrium, and banks earn a strictly positive profit per dollar equal to the multiplier $\nu > 0$.

Part (i) of Proposition 1 describes the optimal allocation for banks lending to *multiple* borrower types. When banks' capacity constraint does not bind (i.e., banks are "unconstrained"), equation (15) holds with $\nu = 0$: competition must bring expected profits to zero, which imposes that the loan rate is just high enough to compensate lenders for the credit risk. The only subtlety is that the credit risk is itself endogenous and in particular depends on the loan rate, so that (15) is actually a fixed-point condition. When banks' capacity constraint binds (i.e., banks are "constrained"), the same logic goes through with a positive multiplier ν. In both the unconstrained and constrained cases, a strict inequality $\frac{\Pi^i}{l^i} > \frac{\Pi^j}{l^j}$ between two types $i \neq j$ would allow banks to increase their total profits by reducing the number of loans n^j to type-j borrowers while increasing the number of loans n^i to type-i borrowers in a way that keeps their capacity constraint satisfied.

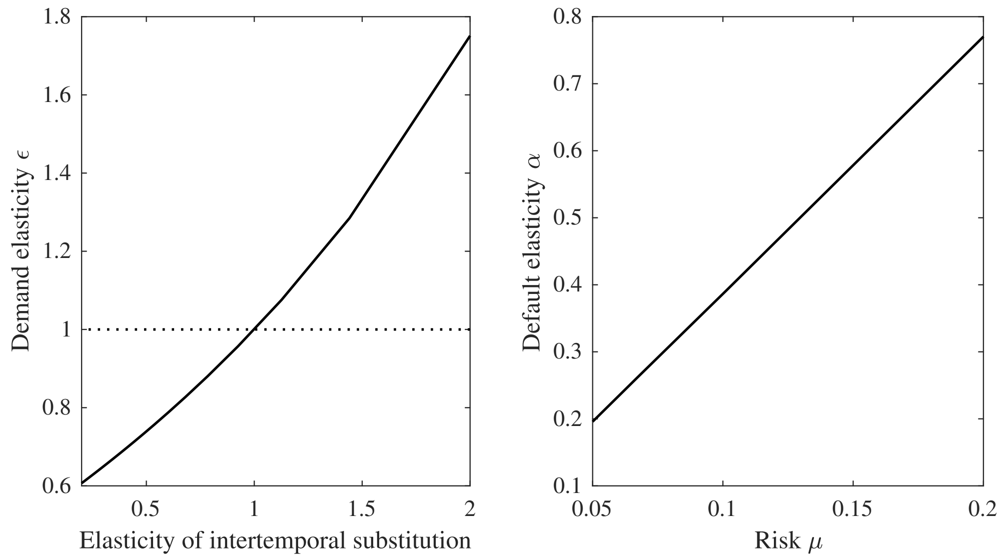


Figure 1: Left panel: Demand elasticity ϵ (solid line) as a function of EIS. Right panel: Default elasticity α as a function of μ.

Part (ii) characterizes which type of equilibrium holds. If banks' capacity constraint is slack, then, as long as a bank makes positive profits $\Pi^i(C_b^i) > 0$, it can increase its profits by deviating to a contract $\tilde{C}_b^i$ offering the same loan size l^i at a slightly lower loan rate $R^i - \epsilon < R^i$. The contract $\tilde{C}_b^i$ is strictly more attractive than the contracts offered by other banks, thus leading type-i borrowers to only apply to bank b. The resulting discrete increase in q_b^i from $1/B$ to 1 would increase total profits $n_b^i q_b^i \Pi_b^i$ and would remain feasible as bank b can always set $\tilde{n}_b^i$ so that this profitable deviation still satisfies the capacity constraint.

2.4 A tale of two elasticities: ϵ and α

Our results rely on two simple statistics. The first one is the demand elasticity ϵ^i, which measures how rate-sensitive borrowers' demand is; the second one is the risk-sensitivity α^i, which measures how credit risk varies with the face value of the loan. Before turning to their implications for credit transmission, we discuss their main determinants.

The value of ϵ^i is driven by the familiar determinants of credit demand elasticities. For instance, in the case of households, lifecycle borrowing is motivated by a consumption-smoothing motive. In this context, the main driver of the elasticity ϵ is the elasticity of intertemporal substitution (EIS). When the EIS is very low, households are unwilling to shift consumption across periods in response to rate changes. When the EIS is high, fluctuations in consumption are less costly and households become more responsive to rate changes.

Figure 1 shows the monotone relationship between ϵ and the EIS in a numerical example.

The dotted line separates relatively inelastic borrowers with an elasticity lower than 1, from relatively elastic borrowers with an elasticity higher than 1. The relationship between the two elasticities is less than one for one because not all consumption variations translate into changes in loan demand. The elasticity of loan demand is exactly equal to one when the elasticity of intertemporal substitution is equal to one too and the substitution and wealth effects cancel out.

Turning to α^i, recall that, following Assumption 1, borrowers' default risk increases with the face value of the loan that they must repay. In principle, there is a distinction between the risk elasticity α^i and the risk itself, as captured by μ^i. In most environments where default is triggered when a latent repayment capacity (income, collateral value, or project return) falls below the promised payment, α^i and μ^i are strongly positively correlated. Intuitively, borrowers who are already close to their default threshold experience a larger change in default probability when F increases further.

Numerically, in all the settings we consider, we confirm that risky borrowers have both a higher and more sensitive default risk in response to higher payments, while safe borrowers have both a lower and less sensitive default risk. We therefore view the positive association between μ^i and α^i as a robust implication of many standard models, not as a knife-edge restriction. It also justifies our terminology: when we speak of "riskier" borrowers below, we have in mind borrowers who are both more likely to default (high μ^i) and whose default probabilities respond more strongly to changes in loan terms (high α^i).[4] This positive relation is also consistent with microeconomic evidence from, e.g., Fuster and Willen (2017), Di Maggio et al. (2017), Gross and Souleles (2002b), Adams et al. (2009), and Einav et al. (2012), showing that in mortgage and consumer credit data, changes in payment obligations or interest rates have much larger effects on delinquency and default among borrowers with high leverage or a low credit score than among safer borrowers.

Figure 1 illustrates this pattern numerically in a different example, where income is distributed uniformly and risk is varied through the bounds of the distribution. We show the default elasticity α^i as a function of borrowers' default rates μ^i as both variables are affected by shifts in borrowers' income distribution. The two measures are again positively correlated: the higher the default risk μ^i of a borrower, the more strongly risk responds to loan terms, which is captured by the elasticity α^i.

Empirical benchmarks. In what follows we use recent empirical work to guide the choice of parameter ranges for the key objects that drive the theory: the interest-rate elasticities of loan demand ϵ^i and the default elasticities α^i.

[4] In special cases, we can derive the positive relation between α^i and μ^i analytically. For instance, in the example described in Section 2.2, the default probability $\mu^i = 1 - (F_{\min}/F)^{\alpha^i}$ is increasing in α^i and the default elasticity is constant equal to α^i. Therefore, μ^i and α^i are positively related.

i. *Demand elasticities.*

In the mortgage market, empirical estimates for the demand elasticity ϵ are generally below 1, ranging from 0.05 in Benetton (2021) (in a structural model of the U.K. mortgage market), to 0.11 in Fuster and Zafar (2021) (in survey data), and 0.5 in Best et al. (2019) (using bunching at LTV thresholds).

Conversely, estimates of demand elasticities in the credit card market are significantly higher: Gross and Souleles (2002a)'s estimates correspond to $\epsilon = 1.3$ while Brauning and Stavins (2025)'s imply ϵ between 1.6 and 2.5.

ii. *Default elasticities.*

To measure the elasticity α, it is convenient to start from the elasticity of μ with respect to the face value, $\frac{Rl\mu'}{\mu}$ which is the object identified in several empirical studies. For instance, Fuster and Willen (2017) estimate $\frac{Rl\mu'}{\mu} = 1.1$ while Di Maggio et al. (2017) find $\frac{Rl\mu'}{\mu} = 2$.

Consistent with observed serious-delinquency rates over similar horizons, a reasonable range for μ is $[0.05, 0.15]$. Taking the midpoint $\mu = 0.10$, together with $\frac{Rl\mu'}{\mu} = 1.5$, implies $\alpha = \frac{\mu}{1-\mu}\frac{R\ell\mu'}{\mu} = 0.17$, while the lower and upper bounds $\mu = 0.05$ or $\mu = 0.15$ would correspond to $\alpha = 0.08$ and $\alpha = 0.26$, respectively.

The values above describe market-wide averages. There is substantial cross-sectional variation, and the available micro evidence implies that these default responses are substantially stronger in riskier segments, consistent with our model and Figure 1. For instance, Fuster and Willen (2017) and Di Maggio et al. (2017) show that payment reductions have particularly large effects on delinquency for borrowers with high loan-to-value ratios or limited liquidity, while Gross and Souleles (2002a), Adams et al. (2009), and Einav et al. (2012) find that borrowing and default responses to changes in interest rates, credit limits, or down-payment requirements are strongest for borrowers who are closer to their constraints. This heterogeneity is precisely what our framework captures by allowing α^i to be increasing in the effective default probability μ^i.

In the language of Geanakoplos (2010)'s "credit surface," α^i can be interpreted as the local slope of the surface in the leverage dimension: it measures how quickly effective default risk and, hence, loan terms tighten as a borrower increases the face value of her debt. Geanakoplos and Rappoport (2019) estimate mortgage and corporate bond credit surfaces and document that credit spreads are increasing and convex in leverage, and more so for high-risk borrowers, consistent with the patterns captured in reduced-form by α^i. Quantitative work such as Diamond and Landvoigt (2022) derives implications of rich credit surfaces in mortgage markets. Our approach, which distills these credit surfaces into simple elasticities α^i, allows us to derive analytical results regarding the interaction between risk and demand elasticities.

3 The transmission of credit supply shocks: a 2×2 framework

In this section, we characterize how shocks to bank funding costs and lending capacity are transmitted to the cross-section of loan contracts. The key mechanism is a *credit-risk multiplier*: because the effective default probability depends on the face value of the loan, changes in the funding cost feed back into default risk and hence into loan rates.

3.1 The interplay between elasticity and risk

The next proposition is our first main result. We derive the heterogenous impact of the two main types of credit supply shocks: changes in banks' funding cost R^f and changes in banks' lending capacity $\overline{L}$.

Proposition 2. *For each borrower type, define the credit-risk multiplier*

$$M^i = \frac{1}{1 - \alpha^i(1 - \epsilon^i)} \tag{18}$$

and the pass-through coefficient η^i

$$\eta^i = M^i \epsilon^i. \tag{19}$$

i. ***Funding cost shocks:*** *Suppose that $\overline{L}$ is high enough that banks are unconstrained initially. The effect of a funding cost shock R^f on type-i borrowers is:*

$$\frac{d \log l^i}{d \log R^f} = -\eta^i, \quad \frac{d \log R^i}{d \log R^f} = M^i \tag{20}$$

ii. ***Lending capacity shocks:*** *Suppose that $\overline{L}$ is low enough that banks are constrained initially. The effect of a change in $\overline{L}$ on type-i borrowers is:*

$$\frac{d \log l^i}{d \log \overline{L}} = \frac{\eta^i}{\eta^{avg}}, \quad \frac{d \log R^i}{d \log \overline{L}} = -\frac{M^i}{\eta^{avg}} \tag{21}$$

where $\omega^i = \frac{N^i \ell^i(R^i)}{\sum N^j \ell^j(R^j)}$ denotes the initial loan share of borrower i and $\eta^{avg} = \sum_j \omega^j \eta^j$ is the average pass-through.

iii. ***Pass-through across risk categories:*** *The pass-through η^i always lies between 1 and the elasticity ϵ^i*

$$\eta^i \in \left[\min\left\{1, \epsilon^i\right\}, \max\left\{1, \epsilon^i\right\}\right], \tag{22}$$

and for two borrower types with the same elasticity $\epsilon^i = \epsilon^j = \epsilon$:

- *If $\epsilon < 1$ (inelastic market), then pass-through is higher for borrowers with* ***higher*** *risk-sensitivity:*

$$\alpha^i > \alpha^j \Rightarrow \eta^i > \eta^j. \tag{23}$$

- *If $\epsilon > 1$ (elastic market), then pass-through is higher for borrowers with* ***lower*** *risk-sensitivity:*

$$\alpha^i > \alpha^j \Rightarrow \eta^i < \eta^j. \tag{24}$$

Proof. See Appendix B. □

Proposition 2 provides closed-form formulas for the endogenous responses of loan contracts in the cross-section of borrowers with different risks and on loan markets with different demand elasticities, in terms of two sufficient statistics. The key object is the credit-risk multiplier M^i, which can be constructed for each borrower by combining the two statistics ϵ^i and α^i. In that sense, the 2×2 patterns in Table 1 are not model-specific, and a large class of settings with endogenous default must display the same comparative statics.

In the case of a funding cost shock $d \log R^f$, total lending summed across borrowers responds with elasticity η^{avg}; in the case of a shock to the lending capacity $\overline{L}$ when banks are constrained, total lending is pinned down by $\overline{L}$ hence pass-through determines the relative transmission to different borrower types. In all cases, the elasticity ϵ^i then translates the quantity response $d \log l^i$ to the price response $d \log R^i$ through $d \log R^i = -\frac{1}{\epsilon^i} d \log l^i$.

When banks are unconstrained, the contract for type i is pinned down by two conditions. First, competition and the zero-profit condition on each dollar lent imply

$$R^i\left(1 - \mu^i(F^i)\right) = R^f. \tag{25}$$

Second, loan sizes must satisfy borrowers' demand $l^i = \ell^i(R^i)$. Taking a proportional change in the funding cost R^f and totally differentiating the zero-profit condition while using the definitions of ϵ^i and α^i yields the rate response

$$\frac{d \log R^i}{d \log R^f} = M^i. \tag{26}$$

and the quantity response (20). The credit-risk multiplier M^i captures how the feedback from loan terms to default risk amplifies or dampens the initial shock. In the special case of exogenous credit risk ($\alpha^i = 0$), the multiplier is $M^i = 1$ and the pass-through coincides with the demand

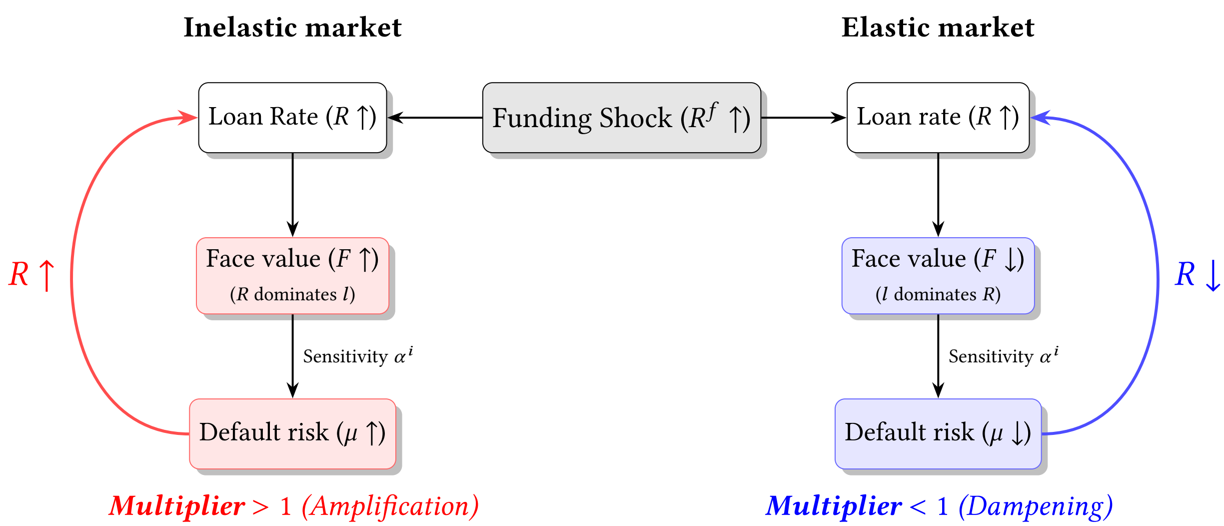


Figure 2: Credit-risk multiplier in inelastic vs. elastic markets.

elasticity ϵ^i. When risk is endogenous ($\alpha^i > 0$), changes in R^i and ℓ^i shift the face value F^i, which moves the effective default probability $\mu^i(F^i)$; the zero-profit condition then requires a further adjustment in R^i, and this feedback is summarized by M^i. When $M_i > 1$, endogenous default risk amplifies the initial shock; when $M_i < 1$, it dampens it.

Figure 2 summarizes this feedback loop. The key observation is that demand elasticities determine whether this feedback loop amplifies or dampens the initial shock, and whether the effect is stronger for riskier or safer borrowers.

In an *inelastic market* ($\epsilon^i < 1$), borrowers adjust quantities little when R^i changes. Following a rise in R^f, the zero-profit condition forces R^i up, while ℓ^i barely moves, so the face value $F^i = R^i\ell^i$ and default risk $\mu^i(F^i)$ increase. Higher default risk then requires an additional increase in R^i to restore zero profit, which *amplifies* the initial funding shock ($M^i > 1$). The stronger the sensitivity of default risk to the face value (higher α^i), the stronger this amplification, so η^i rises with α^i and is pushed up toward 1 from below. Riskier borrowers (high α^i) therefore exhibit larger quantity responses to funding shocks than safer ones, even though their underlying demand elasticity is the same.

In an *elastic market* ($\epsilon^i > 1$), the logic is reversed. A higher R^f again raises R^i, but now borrowers cut quantities so strongly that the face value $F^i = R^i\ell^i$ tends to *fall* when R^i rises. The induced decline in $\mu^i(F^i)$ then works against the funding shock: the zero-profit condition requires a smaller increase in R^i than in a world with exogenous risk ($M^i < 1$). In this case the credit-risk multiplier M^i is below one and *dampens* pass-through. As α^i increases, the dampening becomes stronger, so η^i falls with α^i and is pushed down toward 1 from above. Riskier borrowers (high

α^i) therefore have *smaller* quantity responses to funding shocks than safer borrowers in highly elastic markets.

Remark 2 (Relation to credit rationing literature). *A useful way to interpret Proposition 2 is to contrast it with credit-rationing models in the tradition of Stiglitz and Weiss (1981). A stark difference is that Stiglitz and Weiss (1981) impose fixed loan sizes, which can be understood as a limiting case of our framework where loan demand is perfectly inelastic ($\epsilon \to 0$). This leads to the maximum possible credit-risk multiplier for any given risk-sensitivity $\alpha < 1$. In addition, as the elasticity α increases towards 1 (i.e., the upper bound of the values we allow), the multiplier grows without bounds; going to even higher values $\alpha > 1$ generates the backward-bending profits and extreme rationing they highlight. Our framework shows that the amplification weakens as demand becomes more elastic, and eventually reverses sign at $\epsilon = 1$ (more generally at some threshold $\bar{\epsilon}$, see Section 6.1) hence in inelastic markets a higher α leads to* dampening *instead.*

Figure 3 illustrates how endogenous default risk modifies pass-through numerically.[5] The left panel fixes a low demand elasticity $\epsilon < 1$. The dashed line plots the demand elasticity ϵ while the solid line plots the pass-through η as a function of the default elasticity α. When $\alpha = 0$ (exogenous risk), pass-through coincides with the demand elasticity, $\eta = \epsilon$. As α increases, η rises towards 1: risky borrowers with high α have larger pass-through than safe borrowers, but pass-through remains below unity for all borrowers. The right panel repeats the exercise for a high-elasticity market with $\epsilon > 1$. In that case, η starts again at ϵ when $\alpha = 0$, but now increases in α reduce the pass-through η towards 1. Riskier borrowers (high α) therefore exhibit weaker quantity responses to credit-supply shocks than safe borrowers in highly elastic markets.

The transmission of credit supply shocks to borrowers with different credit risks is heterogeneous across loan markets in the data. For credit cards, bank-level credit expansions tend to be passed through mainly to low-risk (high-FICO) borrowers: exploiting quasi-experimental variation in bank funding, Agarwal et al. (2018) show that most of the extra credit accrues to high-score cardholders, while high-risk cardholders receive little additional credit despite having higher marginal propensities to borrow. For mortgages, by contrast, expansions in credit supply disproportionately relaxed constraints for higher-risk borrowers during the subprime boom. (Mian and Sufi, 2009) document that U.S. ZIP codes with initially low credit scores and incomes experienced the largest relative growth in mortgage credit and the largest subsequent increase in default rates, consistent with credit supply shifting toward riskier households. These contrasting patterns are difficult to reconcile in standard models with exogenous default risk, but they follow directly from our characterization once we allow ϵ^i and α^i to differ across markets in empirically plausible ways.

[5] In order to illustrate the general mechanisms as clearly as possible, we let the default elasticity α vary over a wide range (between 0 and 0.8), which goes somewhat beyond the empirical benchmarks in Section 2.4.

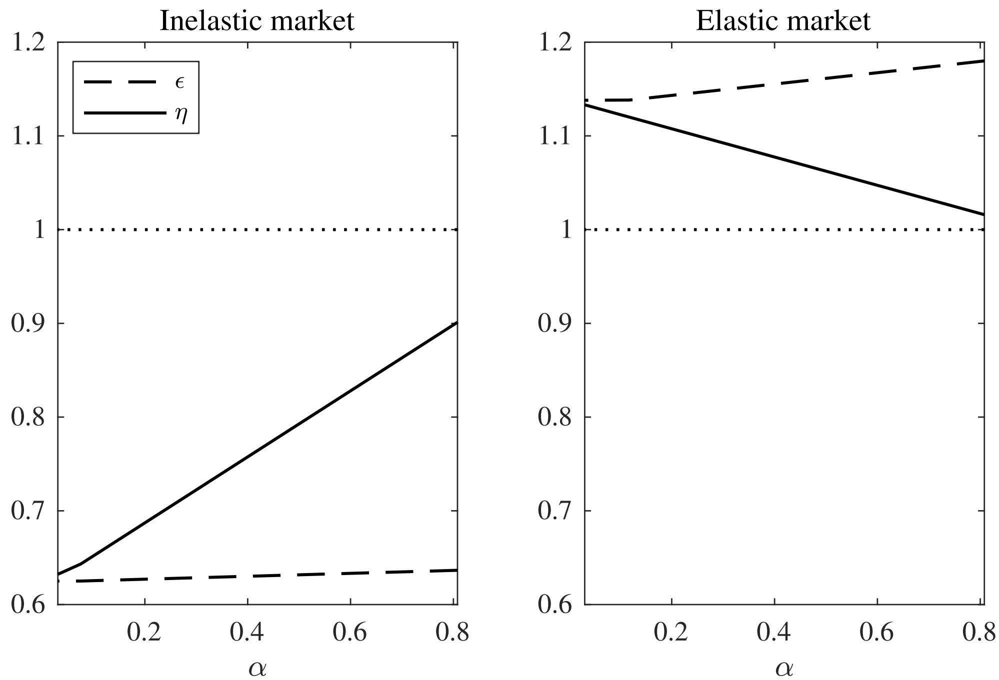


Figure 3: Pass-through η and demand elasticity ϵ as a function of α. Left panel: Inelastic market ($\epsilon < 1$); right panel: elastic market ($\epsilon > 1$).

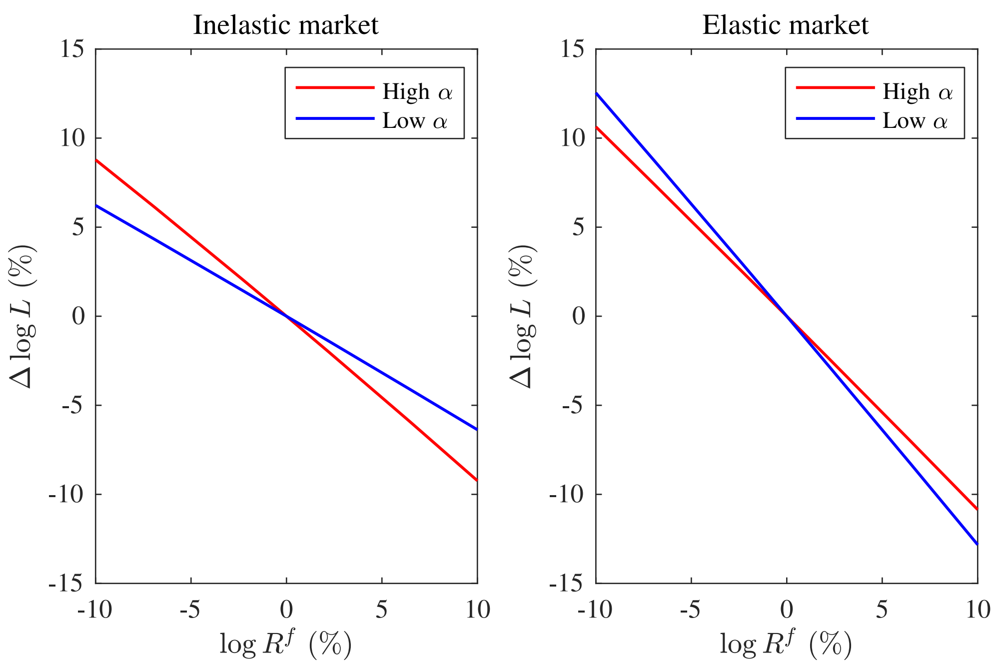


Figure 4: Quantities l^i as a function of R^f for high-α and low-α borrowers. Left panel: Inelastic market ($\epsilon < 1$); right panel: elastic market ($\epsilon > 1$).

Shocks to the risk-free rate R^f, for instance driven by monetary policy, affect banks' cost of funds, and the transmission of these changes display substantial heterogeneity across borrower types. Figure 4 illustrates the implications of Proposition 2 for funding cost shocks. The left panel considers an inelastic market ($\epsilon < 1$). As the funding cost R^f increases, both high-α and low-α borrowers experience declines in loan size l^i, but the decline is steeper for riskier borrowers (red line). Equivalently, monetary easing (a decline in R^f) generates a disproportionate expansion of credit to riskier borrowers. The right panel shows the opposite pattern holds in elastic markets ($\epsilon > 1$): quantities for safe borrowers respond more strongly to changes in R^f than those for risky borrowers.

The pattern in the left panel echoes the large empirical literature on the "risk-taking channel" of monetary policy. Several studies find that, when short-term rates are low, banks expand lending and tilt their portfolios toward ex-ante riskier borrowers or toward loans with higher ex-post default rates; see, e.g., Borio and Zhu (2012), Jimenez et al. (2014), Dell'Ariccia et al. (2017) and Paligorova and Santos (2017). Our model generates a complementary explanation, in which this pattern arises naturally in relatively inelastic loan markets without any incentives to take excessive risk or "reach for yield" in a low return environment. Instead, in our model the equilibrium expansion is tilted toward the high-η^i borrowers (which are the riskier ones here) exactly because the lower loan rates induced by monetary easing improve their effective repayment the most. In the next section, we turn to the model's implications for default risk.

The right panel formalizes the opposite pattern. When $\epsilon > 1$, safe borrowers have a higher pass-through η^i than risky borrowers, so funding shocks induce a reallocation of lending toward safer types. This configuration matches evidence from high-elasticity markets in which credit expansions are tilted toward safer borrowers. In the credit card market, for instance, Agarwal et al. (2018) show that bank funding shocks mainly relax limits for high-credit-score consumers, even though lower-score consumers have higher marginal propensities to borrow. Similar credit reallocation toward safer firms has been documented in some corporate loan settings when banks face funding or regulatory shocks, consistent with $\mathbf{Cov}(\mu_i, \eta_i) < 0$ in those markets. In our framework, both sets of findings arise from the same mechanism once we allow demand elasticities and default elasticities to differ across markets.

3.2 Bank diversification and cross-market spillovers

In this section, we derive implications for cross-borrower effects when banks are diversified across loan markets.

When banks are unconstrained, there is no linkage between different borrower types as banks compete for each type of borrowers up to the point where they make zero profits on each type.

The equilibrium loan contract can be obtained separately for each type, and does not depend on which markets the bank is active on.

However, in equilibria with constrained banks, the optimal allocation of scarce lending capacity to different types of borrowers according to Proposition (1) creates strong linkages between types. At any point in time, there are spillover effects, whereby the effect of an aggregate supply shock on type-i borrowers depends on which type-j borrowers the bank also lends to.

We now use the characterization in Proposition 2 to study how bank diversification across markets shapes the static incidence of credit supply shocks. Consider two markets, indexed by $i \in \{M, C\}$ (abstracting from heterogeneity in risk within each market), with different demand and risk elasticities: a low-elasticity market $\epsilon^M < 1$ (e.g. mortgages) and a high-elasticity market $\epsilon^C > 1$ (e.g. credit cards). For each market we can construct the pass-through coefficient η^i and the average pass-through $\eta^{avg} = \sum_j \omega^j \eta^j$, where ω^j is the share of market j in banks' loan portfolios.

Suppose that each market is served by a different set of *specialized* intermediaries that only lend in that single market. A given shock to intermediary lending capacity or funding costs will mechanically be passed through one-for-one to each market. For instance, if all banks are specialized must tighten lending by 10%, then lending must fall by 10% on each market.

Alternatively, suppose that banks are *diversified* and serve both markets. Proposition 2 shows that a one-time change in lending capacity $\bar{L}$ (in a regime with constrained banks) is distributed across markets according to

$$\frac{d \log \ell^i}{d \log \bar{L}} = \frac{\eta^i}{\eta^{avg}}. \tag{27}$$

Aggregate lending must fall by, say, 10%. But borrowers on markets with a larger pass-through η^i absorb a disproportionately larger share of the marginal change in aggregate lending. For instance, in our numerical examples, the elastic credit card market has a pass-through η^C significantly above one, while the inelastic mortgage market has $\eta^M < 1$. Therefore, relative to a benchmark with specialized banks, bank diversification amplifies the impact of negative credit supply shocks on elastic markets but reduces their impact on inelastic markets. Conversely, a positive shock expands lending more strongly on that market.

Figure 5 illustrates this pattern by showing the transmission of a given aggregate lending capacity shock (a fall in $\bar{L}$) to heterogeneous markets under different scenarios. The left panel considers *specialized* banks: one set of banks only lends on market M (low ϵ) and another only on market C (high ϵ). The right panel considers *diversified* (or "generalist") banks that operate on both markets and face a single lending-capacity constraint. The same shock now reduces lending on both markets with a relative impact governed by the ratio η^i/η^{avg}. Because $\eta^C > \eta^M$, credit card lending contracts more than mortgage lending in percentage terms.

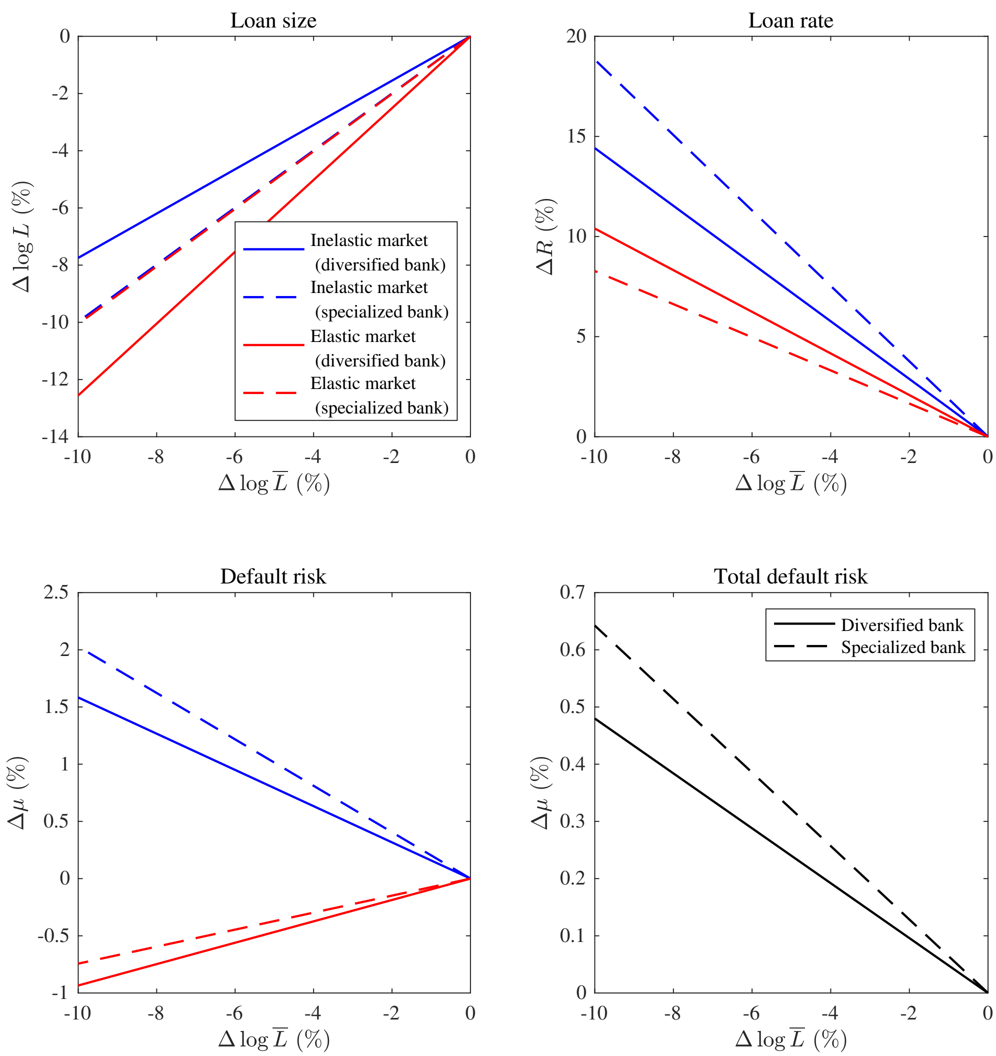


Figure 5: Impact of credit supply shock: specialized banks (dashed lines) vs. diversified banks (solid lines).

Contagion across markets. When the same intermediary operates in multiple segments, credit supply shocks originating in one market transmit to others. Any shock that raises the shadow value of lending capacity, ν, tightens credit conditions in *all* the markets it serves. This generates a form of contagion even when borrower fundamentals are unchanged: spreads rise and quantities contract outside the originating segment because equilibrium pricing satisfies $R^i(1-\mu^i) = R^f + \nu$ for all active contracts. The relative incidence across segments is again governed by the pass-through coefficients η^i: segments with higher η^i absorb a larger share of a given tightening in total capacity.

One salient example is how shocks to regulatory constraints or securitization conditions in one segment propagate to others. In many applications, bank balance-sheet constraints are risk-weighted, with weights ρ^i that differ across loan categories. This generalization is immediate in our framework: the capacity constraint becomes

$$\sum_i \rho^i x^i l^i \leq \bar{L}. \tag{28}$$

The constrained pricing condition (15) generalizes to

$$R^i \left[1 - \mu^i(R^i l^i)\right] - R^f = \rho^i \nu. \tag{29}$$

Changes in ρ^i therefore operate as segment-specific supply shifters, and, when banks are diversified and constrained, induce spillovers to other segments through the endogenous adjustment of ν.

Figure 6 illustrates cross-market contagion when a bank is diversified across two markets, one inelastic and the other elastic. We analyze a shock to the risk weight of one type of borrowers (here the inelastic borrowers, but results are similar if the shock affects the risk weight of elastic borrowers instead). Although the shock originates in one segment, the common balance sheet constraint transmits it to the other. Inelastic borrowers bear a weaker loan size contraction and a stronger rate increase, and experience rising default risk. In the elastic market, even though there is no direct change in risk weights, borrowers face an even stronger reduction in loan size, while their default risk actually declines.

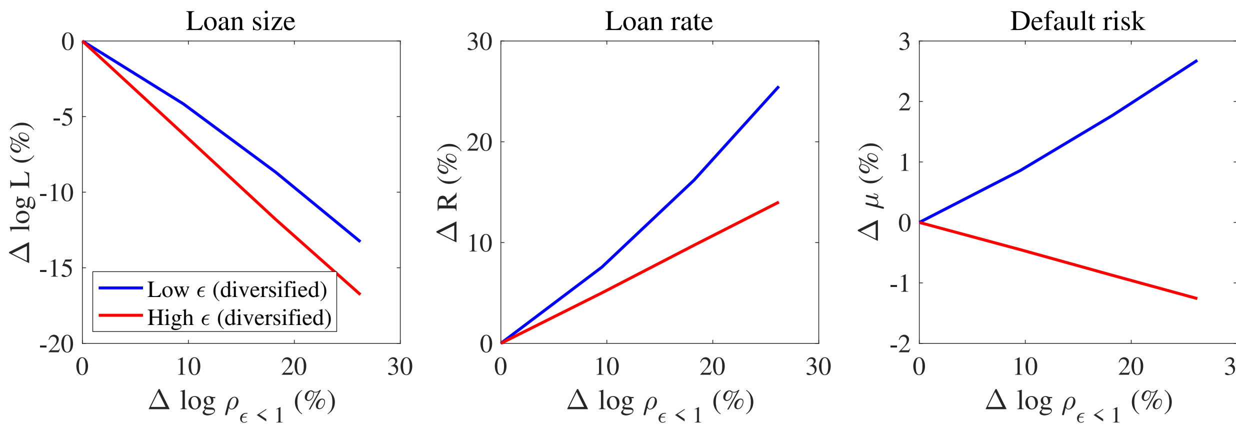


Figure 6: Responses of loan size, loan rate and risk for high-ϵ (red) and low-ϵ (blue) borrowers in response to a shock to ρ for low-ϵ borrowers.

4 Bank risk-taking: cross-sectional estimates and aggregate effects

In this section, we analyze how shocks to the funding cost R^f affect credit risk, both at the borrower level and at the bank portfolio level.[6] As discussed in connection with Figure 4, a large empirical literature documents that monetary policy shocks are associated with systematic changes in the composition of bank lending. When policy rates are low and funding is abundant, banks expand lending and tilt their portfolios toward ex-ante riskier borrowers or toward loans with higher ex-post default rates; when conditions tighten, lending to those borrowers contracts more strongly and portfolios tilt toward safer, larger, or better-collateralized borrowers. These patterns are often described as a "risk-taking channel" of monetary policy on the way down and "flight to quality" on the way up, and they are interpreted as evidence on how the redistribution of lending across borrowers with different risk profiles affects banks' total risk.

Our framework provides a simple laboratory to reconnect these cross-sectional estimates to *aggregate* bank risk. Two features are central. First, default is endogenous and contract-dependent. Second, funding shocks have heterogeneous pass-through to quantities (in part due to contract-dependent risk), hence a common credit supply shock generates non-trivial reallocations of lending across borrower types.

We show that the interaction between these two forces create subtle effects of credit supply shocks on banks' ultimate portfolio risk. First, from Proposition 2, a funding cost shock induces the quantity response $\Delta \log l^i = -\eta^i \Delta \log R^f$. Empirically, cross-sectional regressions relate loan growth to ex-ante measures of borrower or loan risk, therefore estimating whether high-

[6]We focus on funding cost shocks to better relate to the empirical literature, but shocks to the lending capacity $\bar{L}$ would have similar implications.

risk borrowers have higher pass-through η^i. In our notation, for a given common shock $\Delta \log R^f$, such regressions identify the sign and magnitude of

$$\mathbf{Cov}\left(\mu^i, \Delta \log l^i\right) \;=\; \mathbf{Cov}\left(\mu^i, \eta^i\right) \times (-\Delta \log R^f). \tag{30}$$

We can interpret the cross-sectional risk-taking and flight-to-quality patterns in the data as evidence on the *composition* term $\mathbf{Cov}\left(\mu^i, \eta^i\right)$ that will appear below in our expression for aggregate risk. A stronger credit expansion toward riskier borrower corresponds to $\mathbf{Cov}\left(\mu^i, \eta^i\right) > 0$.

The cross-sectional differences captured by (30) are not the full story, however. In our model, any reallocation of lending toward riskier or safer borrowers comes hand in hand with changes in these borrowers' individual default risk.

To study the effect of funding shocks on default risk itself, we decompose the response of total risk into (i) individual changes in default probabilities driven by changes in loan contract terms, and (ii) composition effects due to the heterogeneous transmission of funding shocks across borrower types studied in Proposition 2.

For any variable x^i defined at the borrower-type level, let $\mathbf{E}[x] = \sum^i \omega^i x^i$ denote its average portfolio-level value, where ω^i is the portfolio share of type-i loans. Similarly, we denote $\mathbf{Cov}(x^i, y^i) = \mathbf{E}[x^i y^i] - \mathbf{E}[x^i]\,\mathbf{E}[y^i]$ the covariance of two variables x^i and y^i.

Proposition 3. *The response of borrower i's individual effective default risk μ^i to a shock to R^f around an initial equilibrium with unconstrained banks is, to first order,*

$$\Delta\mu^i = \alpha^i(1-\mu^i)\frac{\epsilon^i - 1}{1-\alpha^i(1-\epsilon^i)} \times \left(-\Delta \log R^f\right). \tag{31}$$

The response of the bank's total portfolio risk $\mathbf{E}[\mu^i]$ can be decomposed, to first order, as

$$\Delta\,\mathbf{E}[\mu^i] = \mathbf{E}\left[\Delta\mu^i\right] + \mathbf{Cov}\left(\mu^i, \eta^i\right) \times \left(-\Delta \log R^f\right) \tag{32}$$

The demand elasticity ϵ^i determines how individual default risk μ^i responds to changes in the funding cost. In particular, in response to a credit expansion induced by monetary easing $d \log R^f < 0$, individual effective default risk μ^i decreases in an inelastic market, but increases in an elastic market. In both cases, the effect is stronger for riskier (high-α^i) borrowers. Intuitively, a change in R^f affects the face value F^i through both the rate and the quantity margin. Using the demand elasticity ϵ^i, we can write $\Delta \log F^i = -\left(1 - 1/\epsilon^i\right)\eta^i\,\Delta \log R^f$. When demand is inelastic ($\epsilon^i < 1$), loan quantities react little to funding shocks, so a rate cut mostly shows up as a lower promised repayment and hence lower default probabilities. When demand is elastic ($\epsilon^i > 1$), quantities react strongly, and the expansion in l^i dominates, so F^i and μ^i increase when rates fall.

Figure 7 illustrates the borrower-level implications of Proposition 3. The left panel considers

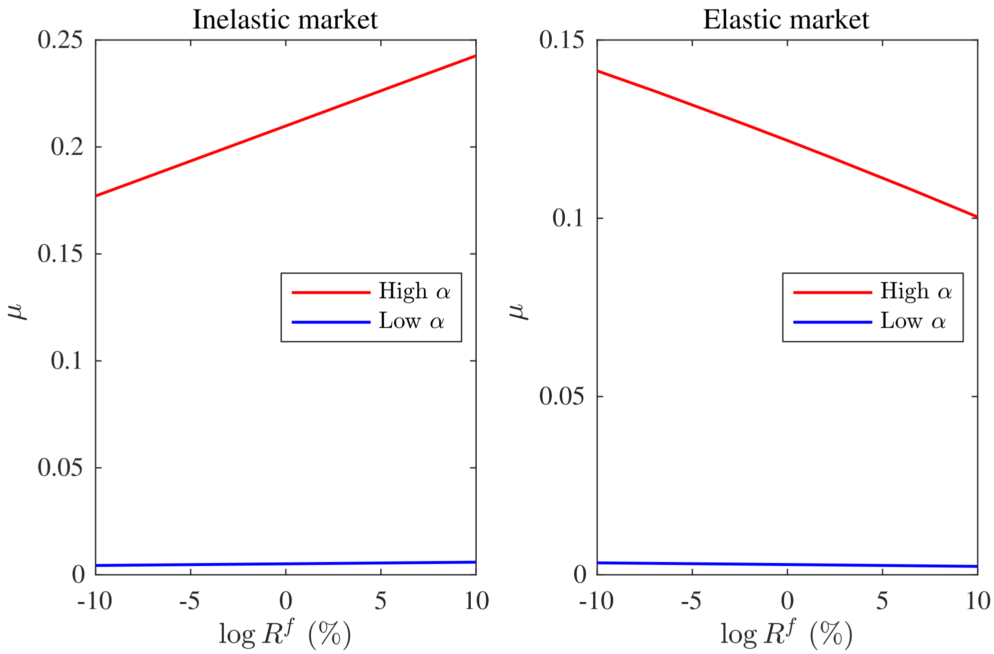


Figure 7: Individual risks μ^i as a function of R^f for high-α and low-α borrowers. Left panel: Inelastic market ($\epsilon < 1$). Right panel: Elastic market ($\epsilon > 1$).

a low-elasticity market with $\epsilon < 1$. As the funding cost R^f increases, equilibrium loan rates rise while loan quantities fall only modestly, so the face value $F^i = R^i l^i$ increases for risky, high-α^i, borrowers and their default probability μ^i rises with R^f. A monetary easing (decline in R^f) therefore reduces individual default risk in inelastic markets, which, by Proposition 2, are precisely the markets in which monetary easing reallocates lending towards the riskier borrowers. Conversely, the right panel considers a high-elasticity market ($\epsilon > 1$), where higher funding costs induce a large contraction in loan quantities. The face value F^i then falls with R^f and default risk μ^i is compressed; accommodative monetary policy raises individual default risk in these markets. This can be viewed as a borrower-level risk-taking channel: in elastic markets the quantity response to funding shocks is strong enough to move promised repayments F^i in the same direction as credit supply.

Equation (32) then decomposes the change in total portfolio risk into two parts. The first term, $E[\Delta\mu^i]$, is a *within-borrower* effect: it captures how individual default probabilities change for a given portfolio composition. The second term, $-\ \mathbf{Cov}(\mu^i, \eta^i)\,\Delta \log R^f$, is a *composition* effect: it captures how a funding shock reallocates lending across borrowers with different risk levels and pass-through coefficients.

When $\mathbf{Cov}\left(\mu^i, \eta^i\right) > 0$, a rate cut ($\Delta \log R^f < 0$) reallocates lending toward borrowers with higher default risk, raising total risk all else equal; when $\mathbf{Cov}\left(\mu^i, \eta^i\right) < 0$, a rate cut reallocates towards safer borrowers. This reallocation component underlies the "risk-taking channel" of

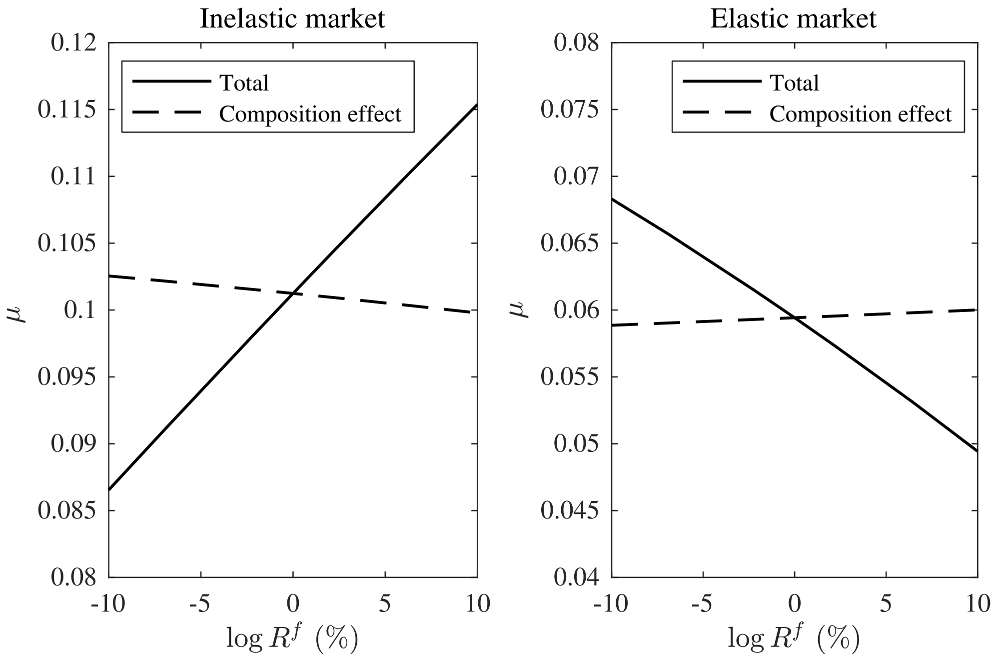


Figure 8: Total risk μ as a function of R^f. Left panel: low elasticity ϵ, right panel: high elasticity ϵ.

monetary policy: we found that in inelastic markets, monetary easing indeed reallocates lending towards riskier borrowers, i.e., $\epsilon < 1$ leads to $\mathbf{Cov}\left(\mu^i, \eta^i\right) > 0$.

Figures 7 and 8 show how these two forces interact and add up. The dashed line in Figure 8 isolates the composition term proportional to $\mathbf{Cov}(\mu^i, \eta^i)$, while the solid line shows how the total portfolio default probability $E[\mu^i]$ changes with $\log R^f$. In the left panel (low-elasticity market), high-μ^i borrowers also have high default elasticity α^i by our microfoundations, and Proposition 2 implies they have higher pass-through η^i. A rate cut therefore reallocates lending toward riskier borrowers, so $\mathbf{Cov}(\mu^i, \eta^i) > 0$ and the dashed line slopes downward in $\log R^f$: holding μ^i fixed, easing tilts the portfolio toward riskier types. At the same time, equation (31) implies that individual default probabilities μ^i fall when R^f falls, and they fall *more* for the high-α^i types. In the calibration underlying Figure 8, this within-borrower effect dominates, so the solid line slopes upward and total portfolio risk *declines* under monetary easing, even though loan growth is strongest for high-risk borrowers.

In the right panel (high-elasticity market), the signs are reversed. When $\epsilon > 1$, Proposition 2 implies that safe borrowers have higher η^i, so $\mathbf{Cov}(\mu^i, \eta^i) < 0$. A rate cut now reallocates lending toward safer borrowers (flight-to-quality composition), and the dashed line slopes upward in $\log R^f$. However, equation (31) implies that individual default probabilities μ^i increase when R^f falls in these markets, and in the calibration this within-borrower effect dominates, so the solid line slopes downward and total portfolio risk *rises* under monetary easing.

The relation between risk levels and risk sensitivities plays a central role in these results. Borrowers with higher default probabilities μ^i also have higher default elasticities α^i. Combined with the pass-through formula $\eta^i = \epsilon^i/[1 + \alpha^i(\epsilon^i - 1)]$, this implies a tight link between risk, default elasticity, and pass-through. Within a given market (fixed ϵ^i), either the rankings of μ^i, α^i, and η^i coincide (when $\epsilon^i < 1$) or they are exactly reversed (when $\epsilon^i > 1$). Equation (31) then shows that $\Delta\mu^i$ is proportional to α^i, so the borrowers who drive the composition effect (those with high η^i when $\epsilon^i < 1$, or low η^i when $\epsilon^i > 1$) are also the borrowers whose individual default probabilities move the most. As a result, we have the following corollary:

Corollary 1. *The within-borrower and composition components of $\Delta E[\mu^i]$ in equation (32) always have opposite signs:*

- *if $\epsilon < 1$ then the within-borrower effect $\mathbf{E}\left[\alpha^i(1-\mu^i)\frac{\epsilon^i-1}{1-\alpha^i(1-\epsilon^i)}\right]$ is negative and the composition effect $\mathbf{Cov}(\mu^i, \eta^i)$ is positive;*
- *if $\epsilon > 1$ then the within-borrower effect $\mathbf{E}\left[\alpha^i(1-\mu^i)\frac{\epsilon^i-1}{1-\alpha^i(1-\epsilon^i)}\right]$ is positive and the composition effect $\mathbf{Cov}(\mu^i, \eta^i)$ is negative.*

The fact that these two components always have opposite signs is a direct consequence of the multiplier structure. The same risk elasticity α^i that makes pass-through higher for riskier borrowers in inelastic markets also makes their individual default probabilities more sensitive, so composition effect and the within-borrower effect have a common fundamental cause. Which of the two dominates is a quantitative question; our numerical examples in Figures 7–8 sit in the region where the within-borrower effect dominates.

The decomposition in equation (32) also clarifies what empirical estimates identify. A regression of $\Delta \log l^i$ on an ex-ante risk proxy for a given policy episode recovers how the pass-through coefficient η^i varies with that risk measure. Cross-sectional estimates are about heterogeneity in η^i, hence reflecting *differences* in default elasticities α^i across borrower types.

By contrast, the within-borrower term $E[\Delta\mu^i]$ in equation (32) is driven by the average sensitivity of default to the contract, summarized by a weighted average of α^i. Aggregating equation (31) across borrowers shows that portfolios loaded on borrowers with high default elasticities display much larger changes in $E[\mu^i]$ for a given movement in R^f than portfolios with uniformly low α^i, even if the cross-sectional dispersion of η^i is similar.[7]

Taken together, equations (31)–(32) and Figures 7–8 highlight two implications for interpreting the risk-taking channel literature. First, the cross-sectional response of lending to a funding

[7] In the calibration underlying Figure 8, the within-borrower dominates the composition effect even though we set $\alpha^i \approx 0$ for the safer borrowers, which maximizes the cross-sectional dispersion of α^i and thus η^i for any given average α^i.

shock depends on the heterogeneity in pass-through η^i (and hence, through Proposition 2, differences in default elasticities α^i) rather than the overall change in portfolio risk. Second, because default is endogenous, the same structure that generates a cross-sectional risk-taking pattern (that is, a positive $\text{Cov}(\mu^i, \eta^i)$ when $\epsilon^i < 1$) simultaneously pushes individual default probabilities in the opposite direction. When this within-borrower effect dominates, monetary easing can lower the overall default risk on bank balance sheets even in environments where loan growth is relatively stronger for high-risk borrowers. Our contribution is to show, in a tractable model, why composition effects alone are only partially informative about the total risk response $\Delta E[\mu^i]$ and how empirical risk-taking estimates can be combined with information on default elasticities to assess the evolution of aggregate bank risk.

5 Pass-through and the dynamics of credit crises

This section embeds the sufficient statistics from Section 3 into a simple dynamic model of intermediary net worth that captures the essence of "financial accelerator" models (e.g., Bernanke et al. 1999, Kiyotaki and Moore 1997, Gertler and Kiyotaki 2010, Brunnermeier and Sannikov 2014). The goal is to characterize how the same pass-through coefficient η that governs static transmission also determines the impact and persistence of credit crises.

5.1 Dynamic environment

Time is discrete and the economy has an infinite horizon. Banks face overlapping generations of borrowers, and their capital varies over time as a function of their profits and losses on loan markets. Starting from a steady state where banks are marginally unconstrained and thus $\nu = 0$, a negative credit supply shock lowers banks' capacity to $\overline{L}_0 < L^*$, i.e., by a proportional amount

$$\delta = \frac{L^* - \overline{L}_0}{L^*} \tag{33}$$

that increases the excess loan premium to $\nu_0 > 0$. After the initial shock the economy evolves deterministically under perfect foresight.

As in the literature on the dynamics of banking crises (Gertler and Kiyotaki, 2010), we assume that banks are unable to raise capital in the short run. Therefore, equity, which determines their lending capacity, recovers slowly through retained earnings. This is the only source of intertemporal linkages in the model: a high excess loan premium at date t increases retained earnings and thus equity growth from t to $t + 1$, which increases lending capacity at $t + 1$. Banks face a leverage constraint

$$\sum_i l_t^i \leq \bar{L}_t = k \cdot n_t \tag{34}$$

at date t where n_t denotes their net worth. We assume that k is low enough that (34) always binds. Banks pay out a fraction $b \in (0, 1)$ of their earnings in every period hence

$$n_{t+1} = n_t(1 + (1 - b)kv_t). \tag{35}$$

As a result, banks' lending capacity, and therefore the total supply of loans, evolves according to

$$\bar{L}_{t+1} = (1 + (1 - b)kv_t)\,\bar{L}_t. \tag{36}$$

Equation (36) shows the standard intermediary asset-pricing mechanism whereby high spreads v_t help recover from negative shocks by recapitalizing the constrained intermediaries. Substituting for the loan contract curve with capacity constraints which arises from the bank's problem and linearizing around the steady state, we obtain the following law of motion for the excess loan premium v_t (i.e., the profit per dollar of loans earned by banks when their lending capacity constraint binds):

$$v_{t+1} = \left(1 - \frac{(1 - b)kR^f}{\eta^{avg}}\right) v_t \tag{37}$$

where η^{avg} is the portfolio-weighted average pass-through coefficient defined in Section 3.

5.2 Implications of pass-through for the impact and persistence of shocks

Proposition 4 describes the impact of the pass-through on the full dynamics of lending.

Proposition 4. *Denote*

$$\varphi = \frac{(1 - b)kR^f}{\eta^{avg}} \tag{38}$$

and the size of the initial credit supply shock $\delta = \frac{L^* - \bar{L}_0}{L^*}$.

(i) Dynamic solution. *In a first-order approximation, the dynamics of the excess loan premium* v_t *and bank lending* L_t *follow*

$$v_t = \frac{R^f}{\eta^{avg}} \delta \left(1 - \varphi\right)^t, \tag{39}$$

$$L_t = L^* \left[1 - \delta \left(1 - \varphi\right)^t\right]. \tag{40}$$

(ii) ***Impact vs. persistence.*** *The cumulative excess loan premium is given by*

$$\sum_{t=0}^{\infty} v_t = \frac{\delta}{(1-b)k} \tag{41}$$

and is therefore independent of the pass-through η^{avg}.

Several properties follow immediately. First, the initial jump in spreads is proportional to the inverse pass-through $1/\eta$. Markets with higher pass-through η experience a smaller initial increase in v_0, hence a milder crisis on impact. Markets with lower η experience a larger jump in spreads.

Second, the speed of recovery is summarized by φ. Therefore, a higher η reduces φ and makes the decay factor $1-\varphi$ closer to one: crises are more persistent. Conversely, a lower pass-through η raises φ and implies a faster return of v_t and L_t to their steady states.

Third, a higher η thus shifts the crisis from "sharp but short-lived" to "mild but long-lasting," while a lower η does the opposite. The pass-through statistic that governed static incidence now generates an intertemporal trade-off between current and future borrowers.

The dependence of η on the underlying elasticities (ϵ, α) ties these dynamic properties directly to demand and risk characteristics. In relatively inelastic markets ($\epsilon < 1$), endogenous default risk ($\alpha > 0$) raises η above the exogenous-risk benchmark ϵ. In these markets, allowing for endogenous risk makes crises milder on impact but more persistent than in a model with exogenous default. In relatively elastic markets ($\epsilon > 1$), the same mechanism lowers η toward one, making crises sharper on impact but less persistent.

The *persistence* of the credit crisis can be measured by the half-life of the excess loan premium v, defined as the time T that it takes for v to revert back to half of its initial value $v_T = v_0/2$, before it ultimately goes back to zero. With Proposition 4, we can compute the half-life of the credit crisis

$$T = \frac{\log 2}{-\log(1-\varphi)}. \tag{42}$$

While η shifts the balance between impact and persistence, the cumulative excess loan premium over time is invariant to pass-through. Part (ii) of Proposition 4 describes this neutrality result, highlighting the benefits from using simple statistics. It shows that, for given bank parameters (b, k) and shock size δ, the cumulative excess premium paid during the crisis is independent of the specific features of credit markets, such as borrower preferences and the microeconomic frictions which generate endogenous default risk $\alpha > 0$. The rich loan-market heterogeneity that matters for static incidence and for the timing of the crisis drops out completely of the present-value measure. Pass-through and default elasticities determine *who* bears the adjustment and *when*, but not the total amount of excess spread that must be generated to recapitalize banks.

5.3 Dynamic effects of bank diversification

We now return to the two-market example from Section 3.2 and compare the dynamics under specialist and generalist banks.

When banks are specialized and each bank lends on a single market, the law of motion in Proposition (4) applies separately on each market with its own pass-through coefficient η^i. A given proportional contraction in lending capacity on market i induces a larger initial increase in spreads and a faster recovery when η^i is small (e.g. on a low-ϵ^i mortgage market), and a smaller initial increase in spreads and a slower recovery when η^i is large (e.g. on a high-ϵ^i credit card market).

When banks are diversified and lend simultaneously on both markets, they face a single lending-capacity constraint and equalize profits per dollar across markets at each date. In this case, the dynamic law of motion is governed by the average pass-through η^{avg}, and the common excess loan premium ν_t drives the joint recovery of lending on both markets.

For a common δ across markets, under bank specialization the average excess loan premium is

$$\bar{\nu}_0^{\text{spec}} = \delta R^f \sum_i \omega^i \frac{1}{\eta^i}. \tag{43}$$

Since $x \mapsto 1/x$ is convex on $(0, \infty)$,

$$\sum_i \omega^i \frac{1}{\eta^i} \geq \frac{1}{\eta^{avg}}, \tag{44}$$

where the right-hand side captures the case of a diversified bank. Thus specialization implies $\bar{\nu}_0^{\text{spec}} > \nu_0^{\text{div}}$, i.e., the *average* initial tightening of credit is stronger than under diversification, holding the mean pass-through constant.

The half-life on market i is

$$T(\eta^i) = \frac{\log 2}{-\log\big(1 - (1-b)kR_f/\eta^i\big)}. \tag{45}$$

For parameter values in the calibration, $T(\eta)$ is increasing and concave in η. Hence, for a fixed mean η^{avg},

$$\mathbf{E}[T(\eta^i)] \ < \ T(\eta^{avg}). \tag{46}$$

With specialized banks, dispersion in pass-through η^i therefore tends to shorten the average half-life across markets, even as it raises the average initial spread. Low-η markets suffer particularly sharp but short crises; high-η markets experience milder but more persistent crises. Proposition 4 applies market by market: for each i,

$$\sum_t v_t^i = \frac{\delta}{(1-b)k}, \tag{47}$$

so the portfolio-weighted average $\sum_t \bar{v}_t^{\text{spec}}$ is again $\frac{\delta}{(1-b)k}$ and does not depend on the distribution of η^i. Heterogeneity reshuffles the crisis across markets and over time, but leaves its present value unchanged.

Dynamic cross-borrower effects. In the case of diversified banks, we have for each borrower type i:

$$\ell_t^i = \ell^{i,*}\left(1 - \frac{\eta^i}{R^f} v_t\right). \tag{48}$$

In our calibration, $\eta^{\text{C}} > \eta^{\text{M}}$, so credit card lending contracts more on impact and remains more depressed for longer than mortgage lending when the same generalist banks serve both markets. Figure 9 illustrates these dynamics by plotting the paths of spreads and loan quantities for the two markets following a common 10% contraction in lending capacity.

The comparison between specialist and generalist banks highlights two mechanisms. First, diversification can generate *contagion*: a shock that primarily affects one segment (e.g. losses on mortgage loans) tightens the common lending-capacity constraint and leads to higher spreads and lower quantities on other segments, even if their fundamentals are unchanged. Second, diversification can also generate *cross-subsidies*: when low-ϵ^i borrowers (with low η^i) pay higher spreads than high-ϵ^i borrowers in the constrained equilibrium, the resulting profits can finance lending to high-ϵ^i borrowers and accelerate the recapitalization of banks that would otherwise be specialized in high-ϵ^i markets. In both cases, the sign and magnitude of cross-market spillovers are governed by the same sufficient statistics (ϵ^i, α^i) that determine static heterogeneous transmission.

6 Extensions

In this section, we discuss how the main results extend to a more general framework. We relax Assumption 1 on the effective default probability, and allow banks to offer loan contracts that are not constrained by the no-rationing condition (10).

6.1 General effective default probability

Our baseline model assumes that the effective default probability μ^i depends only on the face value $F^i = R^i\ell^i$ of the loan. While this assumption delivers considerable tractability and clean

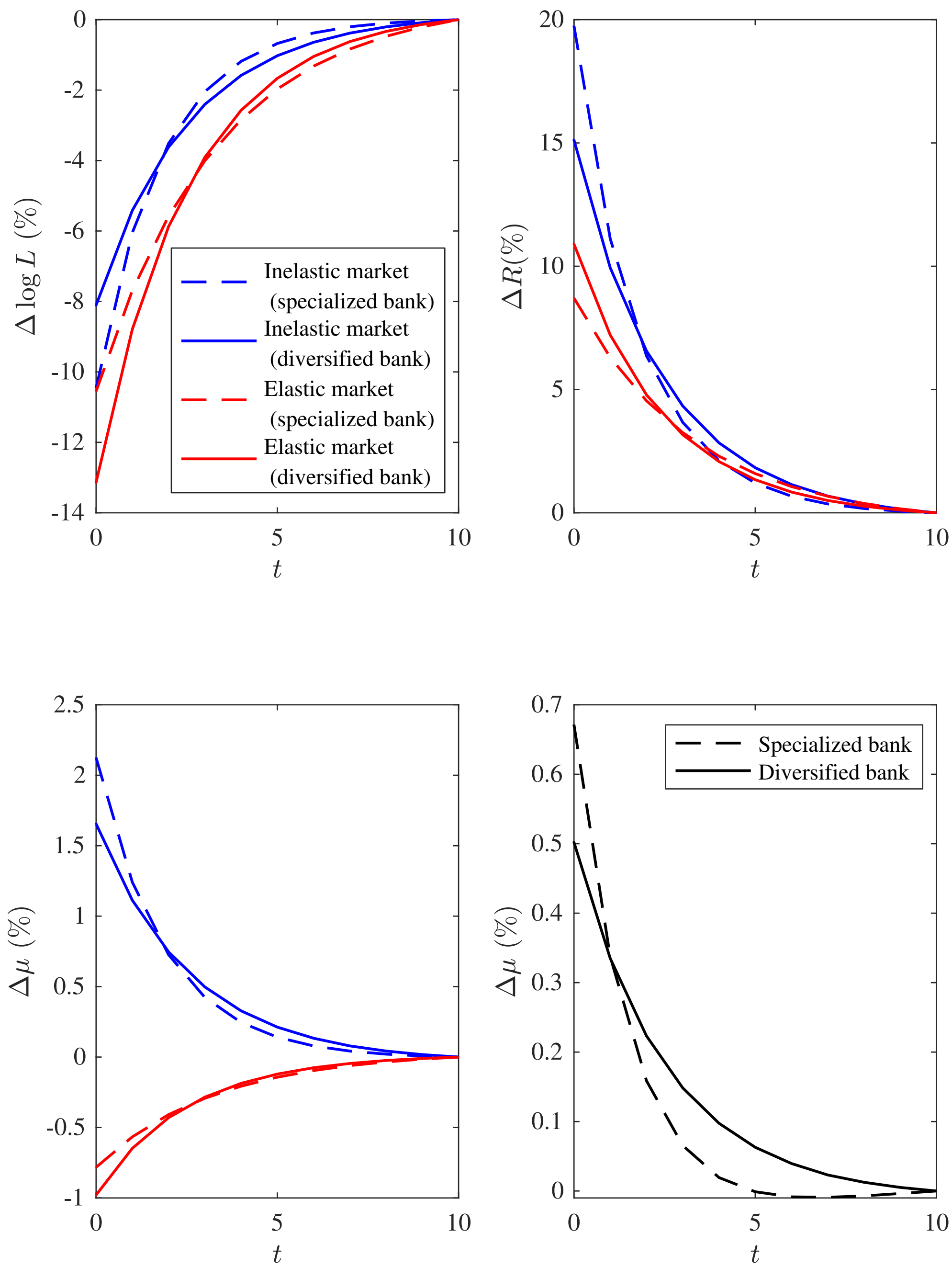


Figure 9: Dynamic impact of credit supply shock: specialized vs. generalist banks.

sufficient statistics, it imposes that rate and quantity effects on default are perfectly linked. We now show that our main results extend to a more general setting where $\mu^i(R^i, l^i)$ depends separately on rates and quantities.

Consider a general effective default probability function $\mu^i(R^i, l^i)$ that can depend separately on the interest rate and loan amount. Define two default elasticities:

$$\alpha_R^i = \frac{R^i \mu_R^i}{1-\mu^i} \quad \text{(default elasticity w.r.t. interest rate)} \tag{49}$$

$$\alpha_l^i = \frac{l^i \mu_l^i}{1-\mu^i} \quad \text{(default elasticity w.r.t. loan size)} \tag{50}$$

The baseline model imposes $\alpha_R^i = \alpha_l^i = \alpha^i$ through Assumption 1. We now allow these elasticities to differ, but maintain tractability by assuming:

Assumption 1'. *For each market, the ratio of default elasticities is constant across borrower types: $\alpha_R^i/\alpha_l^i = \bar{\epsilon}$ for all i in a given market, where $\bar{\epsilon} > 0$ is a market-specific parameter.*

This assumption allows borrowers to differ in their level of default sensitivity to loan size (captured by α_l^i) and loan rates (captured by α_R^i) while maintaining a common structural relationship between these two sources of endogenous default. Borrowers whose risk is more sensitive to loan size also display a higher sensitivity to rates. The baseline model corresponds to the special case $\bar{\epsilon} = 1$. The threshold $\bar{\epsilon}$ is likely higher in markets where a larger loan size enables investment that shifts the distribution of future income upward (e.g., student loans financing human capital or business loans funding productive projects). The effect of l on default is mitigated by improved repayment capacity, whereas a higher loan rate does not have that effect. This makes α_l lower relative to α_R, which in turn increases $\bar{\epsilon}$.

Under this more general specification, the bank's first-order conditions yield a modified pass-through coefficient:

$$\eta^i = \frac{\epsilon^i}{1-\alpha_l^i(\bar{\epsilon}-\epsilon^i)}. \tag{51}$$

The structure of this formula mirrors our baseline expression (19), with the only difference that the threshold that determines heterogeneous transmission is now $\bar{\epsilon}$ rather than unity. As a result, the generalized version of Proposition 2 shows that for two borrower types with the same demand elasticity ϵ but different default sensitivities $\alpha_l^i > \alpha_l^j$ (or equivalently $\alpha_R^i > \alpha_R^j$):

- if $\epsilon < \bar{\epsilon}$ (inelastic market): pass-through is higher for the riskier borrower, $\eta^i > \eta^j$;
- if $\epsilon > \bar{\epsilon}$ (elastic market): pass-through is higher for the safer borrower, $\eta^i < \eta^j$.

Similarly, the generalized specification preserves our qualitative risk-taking results. The change in individual default probability described in Proposition 3 becomes

$$\Delta\mu^i = \alpha_l^i(1-\mu^i)\frac{\epsilon^i - \bar{\epsilon}}{1-\alpha_l^i(\bar{\epsilon}-\epsilon^i)} \times (-\Delta \log R^f) \tag{52}$$

When $\epsilon^i < \bar{\epsilon}$, monetary easing (decline in R^f) reduces individual default risk, precisely in markets where lending shifts toward riskier borrowers. The decomposition into within-borrower and composition effects (32) continues to hold, with the same potential for offsetting effects.

6.2 Intensive-margin credit rationing

In our baseline, we impose a no-rationing constraint (10) for simplicity. We now show that the model yields similar predictions when we relax this assumption, and lenders are allowed to mitigate credit risk by restricting loan size below what borrowers would optimally demand given the equilibrium loan rate.

Recall that we define the *credit wedge* $\tau^i(C^i) = -\frac{l^i V_l^i}{R^i V_R^i}$ to measure how constrained type-i borrowers are if they accept the loan contract C^i. The baseline model imposes $\tau^i = 0$. When intensive-margin credit rationing $\tau^i > 0$ is allowed, equilibrium credit wedges must satisfy

$$\tau^i(R^i, l^i) = \frac{\alpha^i(F^i)}{1-\alpha^i(F^i)}. \tag{53}$$

Equation (53) clarifies that it is the *default elasticity* α^i, rather than the level of default risk μ^i, that governs the tightness of non-price terms. If $\alpha^i = 0$, even if default risk is high, it does not vary with the face value of debt and the equilibrium wedge is $\tau^i = 0$: the allocation can be implemented by simple price posting. When $\alpha^i > 0$, larger loan amounts raise expected losses at the margin, and lenders optimally impose a positive wedge $\tau^i > 0$ (tighten quantity limits) to mitigate endogenous credit risk.

In Appendix C.2, we propose and analyze a measure of effective credit conditions when rationing is allowed. Loan rates are not sufficient to measure credit tightness, because, for instance, it could be optimal to charge risky borrowers with high α^i a *lower* contractual rate R^i than to safer borrowers, combined with a very tight credit limit.

What are the implications of rationing for the pass-through of credit supply shocks? Combining (53) with the borrower's optimality conditions defines an "effective" loan demand curve $\hat{\ell}^i(R)$ that incorporates the equilibrium credit wedge. The comparative statics derived in Section 3 go through unchanged once the standard unrationed demand curve ℓ^i is replaced by $\hat{\ell}^i$. Rationing primarily shifts the level of the wedge, while the elasticity $\hat{\epsilon}^i$ of $\hat{\ell}^i$ remains close to the elasticity ϵ^i of ℓ^i, making the pass-through formulas based on ϵ^i an accurate approximation.

7 Conclusion

We build a model of loan pricing with endogenous default risk and intermediaries with limited lending capacity. Our results provide a unified explanation for key empirical patterns such as the heterogeneous transmission of credit supply shocks across loan markets and borrower risk categories.

A practical implication of the framework is that heterogeneous credit transmission can be organized around two local sufficient statistics measurable in micro data: the interest-rate elasticity of loan demand and the elasticity of effective default risk with respect to the face value of debt. These statistics jointly pin down the pass-through coefficient η, which determines both the incidence of credit supply shocks across borrower types and the mapping from cross-sectional risk-taking estimates to aggregate portfolio risk. In this sense, composition-based evidence alone is not sufficient to infer changes in total bank risk without information on contract-dependent default elasticities.

Beyond our results in this paper, the same approach could be extended to analyze more complex market structures for loans such as imperfect bank competition and borrowers' search for loan terms, which are important extensions for future research.

References

Adams, William, Liran Einav, and Jonathan Levin (2009) "Liquidity Constraints and Imperfect Information in Subprime Lending," *American Economic Review*, Vol. 99, pp. 49–84.

Agarwal, Sumit, Souphala Chomsisengphet, Neale Mahoney, and Johannes Stroebel (2018) "Do Banks Pass through Credit Expansions to Consumers Who want to Borrow?" *The Quarterly Journal of Economics*, Vol. 133, pp. 129–190.

Benetton, Matteo (2021) "Leverage Regulation and Market Structure: A Structural Model of the U.K. Mortgage Market," *The Journal of Finance*, Vol. 76, pp. 2997–3053.

Bernanke, Ben S., Mark Gertler, and Simon Gilchrist (1999) "Chapter 21 The financial accelerator in a quantitative business cycle framework," Vol. 1 of Handbook of Macroeconomics: Elsevier, pp. 1341–1393.

Best, Michael Carlos, James S Cloyne, Ethan Ilzetzki, and Henrik J Kleven (2019) "Estimating the Elasticity of Intertemporal Substitution Using Mortgage Notches," *The Review of Economic Studies*, Vol. 87, pp. 656–690.

Blickle, Kristian, Cecilia Parlatore, and Anthony Saunders (2023) "Specialization in Banking," Working Paper 31077, National Bureau of Economic Research.

Borio, Claudio and Haibin Zhu (2012) "Capital regulation, risk-taking and monetary policy: A missing link in the transmission mechanism?" *Journal of Financial Stability*, Vol. 8, pp. 236–251.

Brauning, Falk and Joanna Stavins (2025) "The Credit Card Spending Channel of Monetary Policy: Micro Evidence from Account-level Data," Working Papers 25-10, Federal Reserve Bank of Boston.

Brunnermeier, Markus K. and Yuliy Sannikov (2014) "A Macroeconomic Model with a Financial Sector," *American Economic Review*, Vol. 104, pp. 379–421.

Bruno, Valentina and Hyun Song Shin (2015) "Capital flows and the risk-taking channel of monetary policy," *Journal of Monetary Economics*, Vol. 71, pp. 119–132.

Buchak, Greg, Gregor Matvos, Tomasz Piskorski, and Amit Seru (2024) "Beyond the Balance Sheet Model of Banking: Implications for Bank Regulation and Monetary Policy," *Journal of Political Economy*, Vol. 132.

Chatterjee, Satyajit, Dean Corbae, Makoto Nakajima, and Jose Victor Rios-Rull (2007) "A Quantitative Theory of Unsecured Consumer Credit with Risk of Default," *Econometrica*, Vol. 75, pp. 1525–1589.

Dell'Ariccia, Giovanni, Deniz Igan, and Luc Laeven (2012) "Credit Booms and Lending Standards: Evidence from the Subprime Mortgage Market," *Journal of Money, Credit and Banking*, Vol. 44, pp. 367–384.

Dell'Ariccia, Giovanni, Luc Laeven, and Gustavo A. Suarez (2017) "Bank Leverage and Monetary Policy's Risk-Taking Channel: Evidence from the United States," *The Journal of Finance*, Vol. 72, pp. 613–654.

Demyanyk, Yuliya and Otto Van Hemert (2009) "Understanding the Subprime Mortgage Crisis," *The Review of Financial Studies*, Vol. 24, pp. 1848–1880.

Di Maggio, Marco, Amir Kermani, Benjamin J. Keys, Tomasz Piskorski, Rodney Ramcharan, Amit Seru, and Vincent Yao (2017) "Interest Rate Pass-Through: Mortgage Rates, Household Consumption, and Voluntary Deleveraging," *American Economic Review*, Vol. 107, pp. 3550–3588.

Diamond, William and Tim Landvoigt (2022) "Credit cycles with market-based household leverage," *Journal of Financial Economics*, Vol. 146, pp. 726–753.

Einav, Liran, Mark Jenkins, and Jonathan Levin (2012) "Contract Pricing in Consumer Credit Markets," *Econometrica*, Vol. 80, pp. 1387–1432.

Fuster, Andreas and Paul S. Willen (2017) "Payment Size, Negative Equity, and Mortgage Default," *American Economic Journal: Economic Policy*, Vol. 9, pp. 167–191.

Fuster, Andreas and Basit Zafar (2021) "The Sensitivity of Housing Demand to Financing Conditions: Evidence from a Survey," *American Economic Journal: Economic Policy*, Vol. 13, pp. 231–65.

Geanakoplos, John (2010) *The Leverage Cycle*, pp. 1–65: University of Chicago Press.

Geanakoplos, John and David Rappoport (2019) "Credit Surfaces, Economic Activity, and Monetary Policy," working paper.

Gertler, Mark and Nobuhiro Kiyotaki (2010) "Financial Intermediation and Credit Policy in Business Cycle Analysis," in *Handbook of Monetary Economics*, Vol. 3, Chap. 11.

Gromb, Denis and Dimitri Vayanos (2018) "The Dynamics of Financially Constrained Arbitrage," *The Journal of Finance*, Vol. 73, pp. 1713–1750.

Gross, David B. and Nicholas S. Souleles (2002a) "Do Liquidity Constraints and Interest Rates Matter for Consumer Behavior? Evidence from Credit Card Data," *The Quarterly Journal of Economics*, Vol. 117, pp. 149–185.

——— (2002b) "An Empirical Analysis of Personal Bankruptcy and Delinquency," *The Review of Financial Studies*, Vol. 15, pp. 319–347.

He, Zhiguo and Arvind Krishnamurthy (2013) "Intermediary Asset Pricing," *American Economic Review*, Vol. 103, pp. 732–70.

Holmstrom, Bengt and Jean Tirole (1997) "Financial Intermediation, Loanable Funds, and The Real Sector," *The Quarterly Journal of Economics*, Vol. 112, pp. 663–691.

Ioannidou, Vasso, Steven Ongena, and José-Luis Peydró (2014) "Monetary Policy, Risk-Taking, and Pricing: Evidence from a Quasi-Natural Experiment*," *Review of Finance*, Vol. 19, pp. 95–144.

Jimenez, Gabriel, Steven Ongena, Jose-Luis Peydró, and Jesus Saurina (2014) "Hazardous Times for Monetary Policy: What Do Twenty-Three Million Bank Loans Say About the Effects of Monetary Policy on Credit Risk-Taking?" *Econometrica*, Vol. 82, pp. 463–505.

Justiniano, Alejandro, Giorgio Primiceri, and Andrea Tambalotti (2019) "Credit Supply and the Housing Boom," *Journal of Political Economy*, Vol. 127, pp. 1317–1350.

Keys, Benjamin J., Tanmoy Mukherjee, Amit Seru, and Vikrant Vig (2010) "Did Securitization Lead to Lax Screening? Evidence from Subprime Loans*," *The Quarterly Journal of Economics*, Vol. 125, pp. 307–362.

Kiyotaki, Nobuhiro and John Moore (1997) "Credit Cycles," *The Journal of Political Economy*, Vol. 105, pp. 211–248.

Livshits, Igor, James MacGee, and Michele Tertilt (2007) "Consumer Bankruptcy: A Fresh Start," *American Economic Review*, Vol. 97, pp. 402–418.

Mian, Atif and Amir Sufi (2009) "The Consequences of Mortgage Credit Expansion: Evidence from the U.S. Mortgage Default Crisis*," *The Quarterly Journal of Economics*, Vol. 124, pp. 1449–1496.

Paligorova, Teodora and Joao A.C. Santos (2017) "Monetary policy and bank risk-taking: Evidence from the corporate loan market," *Journal of Financial Intermediation*, Vol. 30, pp. 35–49.

Paravisini, Daniel, Veronica Rappoport, and Philipp Schnabl (2023) "Specialization in Bank Lending: Evidence from Exporting Firms," *The Journal of Finance*, Vol. 78, pp. 2049–2085.

Robles-Garcia, Claudia (2020) "Competition and Incentives in Mortgage Markets: The Role of Brokers," February.

Stiglitz, Joseph E and Andrew Weiss (1981) "Credit Rationing in Markets with Imperfect Information," *American Economic Review*, Vol. 71, pp. 393–410.

Williamson, Stephen D. (1987) "Costly Monitoring, Loan Contracts, and Equilibrium Credit Rationing," *The Quarterly Journal of Economics*, Vol. 102, pp. 135–145.

Internet Appendix

A Models of endogenous default risk

Income and/or collateral value risk

Consider a general setup with two sources of default risk: future income y_1 and future collateral value K. The joint c.d.f. of y_1 and K is $F(y_1, K)$. For simplicity, we assume that they are independent random variables with c.d.f. $F_y(y_1)$ and $F_K(K)$ and with p.d.f. $f_y(y)$ and $f_K(K)$.

At date $t = 0$, households own an asset with a known value K_0 that we normalize to 0. They borrow l from the bank and consume

$$c_0 = y_0 + l. \tag{54}$$

At the date of repayment $t = 1$, income y_1 and collateral value K are realized. Borrowers have utility

$$\begin{cases} u\left(y_1 - Rl\right) + v(K) & \text{if they repay} \\ u\left(\underline{c}\right) + v(\lambda(1-k)K) & \text{if they default} \end{cases} \tag{55}$$

where $\underline{c}$ denotes a subsistence consumption level available upon default. Hence households default if and only if

$$\begin{aligned} u(y_1 - Rl) + v(K) &\leq u(\underline{c}) + v(\lambda(1-k)K) \\ \Leftrightarrow y_1 &\leq y_1^*(Rl, K) = Rl + u^{-1}\left(u(\underline{c}) + v(\lambda(1-k)K) - v(K)\right) \end{aligned}$$

The partial derivative of the income default threshold with respect to Rl, $y^*_{1,Rl}(Rl, K) = 1$, will be useful below.

This condition captures both liquidity default, stemming from a low realization of income y_1, as well as strategic default, driven by a low realization of the collateral value K. k measures how the collateral value is split between the borrower and the lender in case of default, which captures the pledgeability of the assets. λ is negatively related to the cost of default incurred by the borrower. When λ is low, the borrower retains a low fraction of the split collateral value in case of default, which captures a high cost of defaulting. In terms of state-contingent payoffs, the best case scenario for the borrower is when λ is high and k is low. The best case scenario for the lender is when k is high.

In case of repayment, the bank gets the loan value Rl. In case of default, the bank recovers a fraction k of the collateral value K. Therefore, the effective default risk and its first and second derivatives with respect to the loan value Rl are now:

$$\begin{aligned}\mu(Rl) &= \int_{u(y_1-Rl)+v(K)\le u(\underline{c})+v(\lambda(1-k)K)} (1-\frac{kK}{Rl})dF(y_1,K)\\ &= \int_{K_{min}}^{K_{max}}\left[\int_{y_1\le y_1^*(Rl,K)}(1-\frac{kK}{Rl})dF_y(y_1)\right]dF_K(K)\\ &= \int_{K_{min}}^{K_{max}}(1-\frac{kK}{Rl})F_y(y_1^*(Rl,K))dF_K(K)\end{aligned}$$

Special case: Pareto risk and no recovery value. The default condition can be rewritten as

$$\tilde{g} = g(y_1,K) = y_1 \quad -u^{-1}\left(u(\underline{c}) + v(\lambda(1-k)K) - v(K)\right) \le \quad Rl$$

When the random variable $\tilde{g}$ follows a Pareto distribution F_g with shape parameter α and when the bank does not recover any collateral when the borrower defaults, then the adjusted default risk becomes

$$\mu(Rl) = \int_{\tilde{g}\le Rl} dF_g(\tilde{g}) = F_g(Rl)$$

Then, the elasticity of the adjusted default risk simplifies to a constant:

$$\begin{aligned}\alpha(Rl) &= \frac{\mu'(Rl)Rl}{1-\mu(Rl)}\\ &= \frac{f_g(Rl)Rl}{1-F_g(Rl)}\\ &= \alpha\end{aligned}$$

because $F_g(\tilde{g}) = 1(\tilde{g}\ge g_{min})\left(1-\left(\frac{g_{min}}{\tilde{g}}\right)^{\alpha}\right)$ and $f_g(\tilde{g}) = 1(\tilde{g}\ge g_{min})\frac{\alpha g_{min}^{\alpha}}{\tilde{g}^{\alpha+1}}$.

Adverse selection

When borrowers' types are not observable, a higher repayment Rl attracts a worse distribution of borrowers. This is the classic problem analyzed by Stiglitz and Weiss (1981). For instance, suppose in the previous examples that the distribution of income F varies across types: safe borrowers have a better distribution (in the sense of stochastic dominance) than risky borrowers. In that case a higher face value Rl has the additional effect of attracting relatively more risky borrowers and thus increasing the effective default probability μ.

B Proofs and derivations

Proof of Proposition 1. Fix a symmetric equilibrium and consider a representative bank b, taking other banks' contracts and borrowers' acceptance strategies as given. The bank can equivalently choose the expected number of loans $X_b^i \equiv n_b^i q_b^i$, since for any contract that satisfies borrowers' participation constraint the bank can adjust n_b^i to attain any desired X_b^i.

Given contracts $(R_b^i, \ell^i(R_b^i))$, bank b solves

$$\begin{aligned} \max_{\{X_b^i \geq 0,\, R_b^i\}_{i=1}^I} \quad & \sum_{i=1}^{I} X_b^i\, \ell^i(R_b^i)\Big(R_b^i\big(1-\mu^i(R_b^i\ell^i(R_b^i))\big) - R^f\Big) \\ \text{s.t.} \quad & \sum_{i=1}^{I} \rho^i X_b^i\, \ell^i(R_b^i) \leq L. \end{aligned} \tag{56}$$

Let $\nu \geq 0$ be the multiplier on the capacity constraint. The objective is linear in X_b^i, hence the first-order condition with respect to X_b^i implies that for every type i with $X_b^i > 0$,

$$R_b^i\big(1-\mu^i(R_b^i\ell^i(R_b^i))\big) - R^f = \rho^i \nu. \tag{57}$$

Under the baseline normalization $\rho^i = 1$, this yields equation (15) for all borrower types that receive positive lending in equilibrium. In a symmetric equilibrium with $q_b^i = 1/B$ and market clearing, each type has $X_b^i = N^i/B > 0$, so (15) holds for all i.

Suppose that for some type i, $\Pi^i(C^i) > 0$ in a candidate symmetric equilibrium with slack capacity constraint. Consider a deviation by bank b to a rate $\tilde{R}^i = R^i - \epsilon$ for small $\epsilon > 0$. Since V^i is strictly decreasing in R, type-i borrowers strictly prefer $\tilde{C}^i$, so q_b^i jumps from $1/B$ to 1. By continuity of Π^i in R^i, this deviation is profitable for sufficiently small ϵ, contradicting equilibrium. Hence $\Pi^i = 0$ for all i when the capacity constraint is slack.

If the contracts that solve the zero-profit condition $\Pi^i(\ell^i(R^i), R^i) = 0$ for each i satisfy the capacity constraint (8), then banks are unconstrained and $\nu = 0$ in equilibrium.

If instead (8) fails, then the capacity constraint must bind in equilibrium and $\nu > 0$, and equilibrium contracts satisfy (15) together with market clearing and the capacity constraint.

Proof of Proposition 2. Let

$$x \equiv R^f + \nu, \tag{58}$$

so that the equilibrium pricing condition can be written as

$$R^i\big(1-\mu^i(F^i)\big) = x. \tag{59}$$

Take logs in (59) and totally differentiate:

$$\mathrm{d}\log x = \mathrm{d}\log R^i + \mathrm{d}\log(1 - \mu^i(F^i)) \tag{60}$$

with

$$\mathrm{d}\log(1 - \mu^i(F^i)) = -\alpha^i \,\mathrm{d}\log F^i. \tag{61}$$

Moreover, since $F^i = R^i \ell^i(R^i)$, we have

$$\mathrm{d}\log F^i = \mathrm{d}\log R^i + \mathrm{d}\log \ell^i = \left(1 - \epsilon^i\right)\mathrm{d}\log R^i, \tag{62}$$

where $\epsilon^i \equiv -\frac{R^i}{\ell^i}\frac{\mathrm{d}\ell^i}{\mathrm{d}R^i}$. Combining the above gives

$$\mathrm{d}\log x = \Big(1 - \alpha^i(1 - \epsilon^i)\Big)\mathrm{d}\log R^i. \tag{63}$$

Therefore

$$\mathrm{d}\log R^i = M^i \,\mathrm{d}\log x, \qquad \mathrm{d}\log l^i = -\epsilon^i \,\mathrm{d}\log R^i = -\eta^i \,\mathrm{d}\log x, \tag{64}$$

with $M^i \equiv \left(1 - \alpha^i(1 - \epsilon^i)\right)^{-1}$ and $\eta^i \equiv \epsilon^i M^i$.

Funding cost shocks (unconstrained banks). If banks are unconstrained, $\nu = 0$ and thus $x = R^f$. Equation (64) immediately yields

$$\frac{\mathrm{d}\log R^i}{\mathrm{d}\log R^f} = M^i, \qquad \frac{\mathrm{d}\log l^i}{\mathrm{d}\log R^f} = -\eta^i. \tag{65}$$

Lending-capacity shocks (constrained banks). If banks are constrained, the capacity constraint binds:

$$L = \sum_{i=1}^{I} N^i \,\ell^i(R^i). \tag{66}$$

Log-differentiating (66) gives

$$\mathrm{d}\log L = \sum_i \omega^i \,\mathrm{d}\log l^i, \qquad \omega^i \equiv \frac{N^i \ell^i(R^i)}{\sum_j N^j \ell^j(R^j)}. \tag{67}$$

Using (64),

$$\mathrm{d}\log L = -\Big(\sum_i \omega^i \eta^i\Big)\mathrm{d}\log x = -\eta^{avg} \,\mathrm{d}\log x, \tag{68}$$

where $\eta^{avg} \equiv \sum_i \omega^i \eta^i$. Hence $\mathrm{d}\log x = -\frac{1}{\eta^{avg}}\mathrm{d}\log L$. Substituting back into (64) yields

$$\frac{\mathrm{d}\log \ell^i}{\mathrm{d}\log L} = \frac{\eta^i}{\eta^{avg}}, \qquad \frac{\mathrm{d}\log R^i}{\mathrm{d}\log L} = -\frac{M^i}{\eta^{avg}}. \tag{69}$$

Finally,

$$\frac{\partial \eta^i}{\partial \alpha^i} = \frac{\epsilon^i(1-\epsilon^i)}{\left(1-\alpha^i(1-\epsilon^i)\right)^2}, \tag{70}$$

so η^i is increasing in α^i if $\epsilon^i < 1$ and decreasing in α^i if $\epsilon^i > 1$.

Intensive rationing: With intensive-margin rationing allowed, optimal contracts solve

$$\max_{R^i, l^i} V^i(R^i, l^i) \quad \text{s.t.} \quad R^i\left(1-\mu^i(R^i l^i)\right) = x. \tag{71}$$

Let λ be the multiplier and define $g(R, l) \equiv R(1-\mu(Rl)) - x$. The FOCs imply $V_R + \lambda g_R = 0$ and $V_l + \lambda g_l = 0$, hence $V_l/V_R = g_l/g_R$ with

$$g_R = (1-\mu) - R\mu'(Rl)\, l = (1-\mu)(1-\alpha), \qquad g_l = -R^2\mu'(Rl) = -(1-\mu)\frac{R\alpha}{l}. \tag{72}$$

Therefore

$$\tau = -\frac{lV_l}{RV_R} = -\frac{l}{R}\frac{g_l}{g_R} = \frac{\alpha}{1-\alpha}. \tag{73}$$

Proof of Proposition 3. With unconstrained banks, Proposition 2 implies $d\log \ell^i = -\eta^i\, d\log R^f$ and $d\log R^i = M^i\, d\log R^f$. Hence the face value $F^i \equiv R^i\ell^i$ satisfies

$$d\log F^i = d\log R^i + d\log \ell^i = (M^i - \eta^i)\, d\log R^f = M^i(1-\epsilon^i)\, d\log R^f. \tag{74}$$

Using $\eta^i = M^i\epsilon^i$, this can be written as $d\log F^i = -(1-1/\epsilon^i)\eta^i\, d\log R^f$. By definition of α^i,

$$d\mu^i = \mu^i_F\, dF^i = (1-\mu^i)\alpha^i\, d\log F^i, \tag{75}$$

which yields equation (31) in the proposition.

For total portfolio risk, write $E[\mu^i] = \sum_i \omega^i\mu^i$. A first-order expansion gives $\Delta E[\mu^i] = E[\Delta\mu^i] + \mathbf{Cov}(\mu^i, \Delta\log \ell^i)$. Since $\Delta \log \ell^i = -\eta^i\, \Delta\log R^f$, the composition term becomes $\mathbf{Cov}(\mu^i, \eta^i) \times (-\Delta\log R^f)$.

Proof of Corollary 1. Fix a market with common demand elasticity ϵ. Equation (31) implies that $\Delta\mu^i$ is proportional to $\alpha^i(\epsilon-1)$, so the within-borrower component $E[\Delta\mu^i]$ has the sign of $(\epsilon-1)$. Moreover, $\eta^i = \epsilon/[1-\alpha^i(1-\epsilon)]$ is increasing in α^i when $\epsilon < 1$ and decreasing in α^i when $\epsilon > 1$. Under the maintained positive association between μ^i and α^i, it follows that $\mathbf{Cov}(\mu^i, \eta^i)$ is positive when $\epsilon < 1$ and negative when $\epsilon > 1$. Therefore the composition component $\mathbf{Cov}(\mu^i, \eta^i) \times (-\Delta\log R^f)$ has the opposite sign of $E[\Delta\mu^i]$ in both cases.

Proof of Proposition 4. Linearizing the static equilibrium mapping around $v = 0$ yields $d \log L = -\eta^{avg} \, d \log(R^f + v)$, so to first order, $L/L^* = 1 - (\eta^{avg}/R^f)v$ and

$$v = (R^f/\eta^{avg})(1 - L/L^*). \tag{76}$$

Because the leverage constraint binds, $L_t = \bar{L}_t$ and $\bar{L}_{t+1} = (1 + (1-b)kv_t)\bar{L}_t$, hence

$$v_{t+1} = \left(1 - \frac{(1-b)kR^f}{\eta^{avg}}\right)v_t. \tag{77}$$

Solving the first-order difference equation yields the result.

Part (ii): Using $v_t = (R^f/\eta^{avg})\delta(1-\varphi)^t$ with $\varphi = (1-b)kR^f/\eta^{avg}$,

$$\sum_{t=0}^{\infty} v_t = \frac{R^f}{\eta^{avg}}\delta \sum_{t=0}^{\infty}(1-\varphi)^t = \frac{R^f}{\eta^{avg}}\delta \cdot \frac{1}{\varphi} = \frac{\delta}{(1-b)k}. \tag{78}$$

C Other Extensions

C.1 Long-term loans

Our static model can also nest the case of long-term debt under mild restrictions on the amortization schedule. Suppose that the borrower obtains l at $t = 0$ and then faces a repayment $x_t l$ at each date $t = 1, \ldots, T$ where T is the maturity of the loan and the coupon rate $x_t \geq 0$ is set at the contracting time $t = 0$.

Let $q_{0,t}$ denote the date-0 price of zero-coupon bond maturing at date t. Then the date-0 present value of repayments absent default is$\sum_{t=1}^{T} q_{0,t} x_t l$. The case of long-term debt can be mapped to the static model by defining R as

$$R = \sum_{t=1}^{T} q_{0,t} x_t, \tag{79}$$

such that the present value of repayments absent default is Rl. Then we can rewrite each period's repayment as $x_t l = \delta_t R l$, where $\sum_{t=1}^{T} q_{0,t}\delta_t = 1$.

Consider first the case of liquidity default in which the borrower defaults at t if and only if the date-t repayment exceeds the date-t income $x_t l > y_t$ or equivalently $\delta_t F > y_t$. Then the ex-ante default probability is

$$\mu = \sum_{t=1}^{T} \mathbf{P}_0 \left[y_s \geq \delta_s F \text{ for all } s < t \text{ and } y_t < \delta_t F \right] \tag{80}$$

which is only a function of $F = Rl$, just like in the static model.

Alternatively, consider the case of strategic default in which the borrower defaults at t if and only if the remaining debt burden date-t exceeds some threshold Z_t, which may depend on the path of income (e.g., through the penalties captured by κ in the static model). The remaining debt burden at some date $t \geq 1$ is $\mathcal{R}_t l$ where

$$\mathcal{R}_t l = F \cdot \sum_{s=t+1}^{T} q_{t,s}\delta_s$$

hence at each point in time, the going-forward incentives to default are still a function of the ex-ante present value of repayments absent default F, exactly as in the static model. Therefore, ex ante, the expected default probability is also a function of F.

C.2 Rationing and effective credit conditions

The presence of endogenous default risk primarily affects the *level* of borrower tightness through the wedge τ^i. This is important empirically because movements in contractual rates alone need not be sufficient to measure how tight credit is for borrowers when non-price terms adjust. This motivates the following effective-rate representation. Suppose that under the equilibrium contract $C^i = (R^i, l^i)$ type-i borrowers face a credit wedge τ^i. There exists an *effective rate* $\hat{R}^i \geq R^i$ such that the borrower is indifferent between the equilibrium contract and unrationed borrowing at rate $\hat{R}^i$:

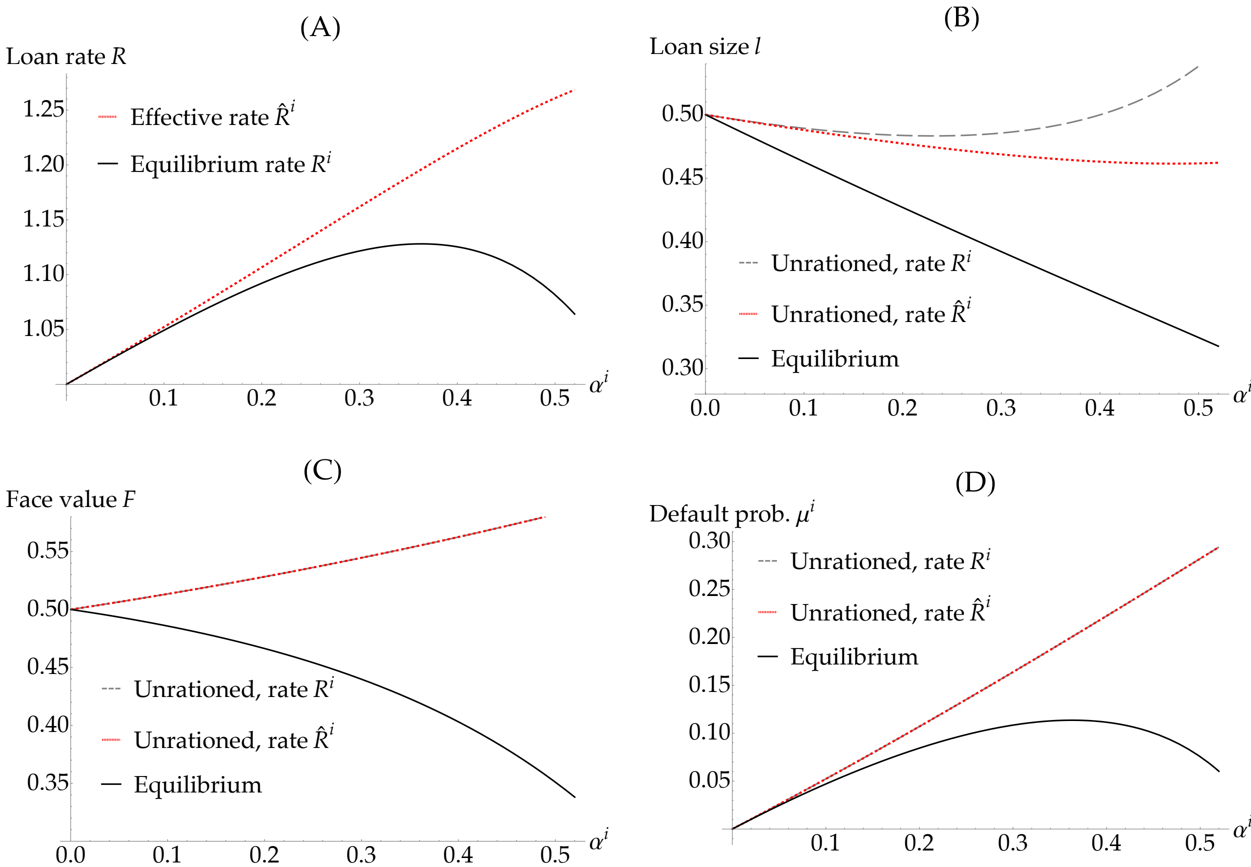
$$V^i(R^i, l^i) = V^i(\hat{R}^i, \ell_u^i(\hat{R}^i)). \tag{81}$$

The effective rate $\hat{R}^i$ is strictly above R^i if and only if $\tau^i > 0$. A positive credit wedge means that the equilibrium contract $C^i = (R^i, l^i)$ is such that $V^i(R^i, l^i) < V^i(R^i, \ell_u^i(R^i))$, where $\ell_u^i(R^i)$ is the unrationed loan demand given rate R^i, defined as the solution to $V_l^i(R^i, l_u^i(R^i)) = 0$. Since $V^i(R^i, \ell_u^i(R^i))$ is a decreasing function of R^i, going from the contractual rate R^i to the effective rate $\hat{R}^i$ replicates the equilibrium welfare through a higher rate $\hat{R}^i$ and a higher loan size $\ell_u^i(\hat{R}^i)$. This is not the equilibrium outcome because even though borrowers are kept indifferent, lender profits are lower.

When non-price terms adjust, the contractual rate R^i can substantially understate how tight credit is for borrowers. In particular, as α^i rises and quantity limits tighten, borrowers can experience a sharp deterioration in effective borrowing conditions even if the contractual rate does not rise; the example below shows that R^i may even fall. The effective rate $\hat{R}^i$ defined in (81), which summarizes both prices and non-price terms, is therefore a more appropriate object for measuring borrower-level credit conditions.

In Figure 10, we consider equilibria with unconstrained banks (i.e., high enough lending capacity $\overline{L}$) to focus on how loan contracts differ in the cross-section of borrowers.

Figure 10: Equilibrium and counterfactual loan contracts as a function of borrower risk.



Panel A: Loan rate as a function of borrower risk α^i (i.e., the Pareto shape parameter of the borrower's income distribution where a higher α^i corresponds to a riskier distribution). The solid black curve depicts the actual equilibrium rate R^i. The dotted red curve depicts the effective rate $\hat{R}^i$ defined in (81).
Panel B: Loan size as a function of borrower risk α^i. The solid black curve depicts the equilibrium loan size l^i. The dashed gray curve depicts the unrationed loan demand at the actual equilibrium rate R^i. The dotted red curve depicts the unrationed loan demand at the effective rate $\hat{R}^i$ defined in (81).
Panel C: Face value as a function of borrower risk α^i. The solid black curve depicts the equilibrium face value $R^i l^i$. The dashed gray curve depicts the unrationed face value at the actual equilibrium rate R^i. The dotted red curve depicts the unrationed face value at the effective rate $\hat{R}^i$ defined in (81).
Panel D: Default probability μ^i as a function of borrower risk α^i. The solid black curve depicts the equilibrium default probability μ^i. The dashed gray curve depicts μ^i evaluated at the unrationed loan demand given the contractual rate R^i. The dotted red curve depicts μ^i evaluated at the unrationed loan demand at the effective rate $\hat{R}^i$ defined in (81).

Panel A illustrates the difference between contractual and effective loan rates. The solid black curve depicts the rate R^i charged to borrowers by banks while the dotted red curve represents the effective rate $\hat{R}^i$. Both are functions of the borrower type captured by the parameter α^i on the x-axis. The default probability and the default elasticity are both increasing in α. The contractual loan rate R^i is non-monotonic in risk α^i: it increases with α^i at low risk levels, but decreases with risk at sufficiently high α^i as the default elasticity becomes so high that increasing the contractual rate would only raise credit risk and lower lenders' profits. As a result, lenders tighten the credit wedge τ^i as α^i grows, as can be seen from the increasing gap between $\hat{R}^i$ and R^i.

High risk borrowers do borrow less and face tighter credit conditions, but this is achieved through a tightening of their quantity limit l^i instead of a higher rate, as shown in Panel B. We contrast the solid black line depicting the equilibrium loan size l^i with a dashed gray line, showing the unrationed loan demand $l_u^i(R^i)$ that would hold at the contractual rate. At high levels of risk α^i, the contractual rate R^i is very low, hence the resulting unrationed loan demand would be very high, but the equilibrium loan size is much lower, which reflects how rationed credit is. For completeness, the dotted red line shows the unrationed loan demand $l_u^i(\hat{R}^i)$ that would arise under the effective rate. These figures show how allowing for a variable loan size is crucial for our results and how they contrast with seminal analyses of extensive-margin credit rationing (Stiglitz and Weiss, 1981).

Panel C shows the face value $F = Rl$ under the different scenarios. The striking difference is that the equilibrium face value F^i decreases sharply with α^i, whereas under unrationed loan demand it increases slightly with α^i. Note that with logarithmic utility and no initial income, the face value under unrationed loan demand only depends on the type α^i , but is independent of whether the interest rate is equal to the equilibrium rate R^i or the effective rate $\hat{R}^i$, hence the dashed gray and dotted red lines coincide. Therefore, the face value increases with risk under unrationed loan demand even though the effective rate $\hat{R}^i$ increases with α^i. By contrast, the face value under the equilibrium contract falls sharply with risk because lenders ration quantities relative to the unrationed loan demand, even in the high-α region where the equilibrium rate R^i becomes decreasing in α^i.

Finally, Panel D shows how the effective default probability μ^i (which is here simply the default probability given the assumption of zero recovery rate upon default) varies with α^i. Under the equilibrium contract, μ^i is non-monotonic, first increasing with α^i but then falling with α^i. The non-monotonicity of the default probability μ^i parallels the non-monotonicity of the equilibrium loan rate R^i since lenders make zero profits. This pattern is surprising because as α^i grows, the default probability μ^i increases for any fixed face value F, which is why we refer throughout to high α^i borrowers as "high risk". However, the equilibrium non-monotonicity of μ^i depicted in the solid black line arises from two offsetting forces. On the one hand, the mechanical increase in the default probability for a given face value tends to push μ^i up as α^i increases; on the other hand, lenders ration credit and the face value so much at high values of α^i that μ^i eventually decreases with α^i.